SIMULATIONS OF NATURAL GAMMA BACKGROUND RADIATION IN BUILDINGS

by

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ABSTRACT

Exposure to natural background radiation in buildings is significant because humans spend a lot of their time in them. In this study, a simulation code was developed from GEANT4 Tool for Particle Simulation (TOPAS) to investigate the dose received by an individual in a room from the gamma energies of natural background radionuclides, that is ^{40}K, and the decay products of ^{238}U and ^{232}Th, emitted from walls. Dose from these simulations was recorded for an individual positioned at the centre of a model room and near the walls. Different room sizes were also simulated to investigate their effect on the dose recorded for the individual. The results demonstrated that ^{40}K contributes the highest dose among the radionuclides, ^{232}Th decay series contributed the most after ^{40}K and ^{238}U decay series contributed the least. The simulations made with the individual near the walls proved that most of the dose an individual receives in a room is from the wall they are close to. Dose received from the centre of the room was therefore lower than dose received closer to the walls. The room was defined with rectangular dimensions. This allowed radiation from a bigger wall and a smaller wall to be simulated. The results showed an increase in the dose received from the smaller walls compared to the bigger ones. For the different room sizes, high doses were recorded in the small room, the model room received lower doses in comparison to the small room and the large room received even less dose than all the other rooms.

Keywords

Natural background radiation, ^{238}U, ^{232}Th, ^{40}K, Radon, gamma radiation, wall thickness, concrete, TOPAS.

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1. INTRODUCTION

Natural background radiation is the most predominant source of radiation to mankind, contributing about 80% of the ionizing radiation that accounts for public exposure in a year (Ramachandran, 2011, Shahbazi-Gahrouei et al., 2013). Even in modern times, where technological advancements have seen the invention of numerous artificial radiation sources like industrial nuclear sources and radiography, these sources contribute less to the overall public exposure dose, mainly because of stringent radiation protection regulations to prevent overexposure. The high contribution of natural background radiation renders exposure to it unavoidable, and to some extent, it is safe to be exposed to it if levels are below recommended reference values. However, the concentration of these natural radiation sources is heavily dependent on the geographical location and the geology of the region, such that, high levels of radiation has been noted in some regions that exceeds recommended threshold values (Sohrabi, 2013, Hendry et al., 2009).

Natural background radiation is grouped into two main categories based on its origins. These are cosmic and terrestrial natural background radiation. Cosmic radiation arises from extra-terrestrial origins related to cosmic ray interactions while terrestrial radiation is present everywhere on earth, from soil to rocks and air and they are believed to have existed even before the creation of earth (Ramachandran, 2011). However, due to the influence of altitude on its exposure contribution, exposure to cosmic rays is less. Most exposure to humans is from the radionuclides of terrestrial natural radiation (Shahbazi-Gahrouei et al., 2013). The most abundant terrestrial natural radionuclides are uranium (^{238}U) decay series, thorium (^{232}Th) decay series and potassium (^{40}K). The decay of ^{238}U and ^{232}Th leads to the production of radon gas, which contributes significantly to the health risks posed by natural radiation sources, specifically in the cause of lung cancer (Barros-Dios et al., 2002, Darby et al., 2005).

The gamma rays produced from the ^{40}K and decay progenies of ^{238}U and ^{232}Th can contribute unwanted doses to individuals indoors. It is estimated that humans spend about 80% of their life indoors (Mehdizadeh et al., 2011). Therefore, investigating these radiations in building materials, to quantify the amount of exposure and implement radiation protection measures that prevents the effects from them is an imperative. However, it is difficult to investigate this manually while accounting for different factors such as different thickness of walls, different room sizes and the introduction of radiation absorbers in a specific region. Monte-Carlo simulations have been proven to produce excellent results in modelling natural background radiation, backed by relevant physics, without the need for any complicated calculations (Li et al., 2016, Hung et al., 2017)

This study aimed at developing a code from Tool for Particle Simulation (TOPAS), a Geant4 Monte-Carlo simulation toolkit (Allison et al., 2016), to investigate the dose received from gamma energies in a model building made of concrete from these natural background radiations. The simulations were done for the gamma energies of the three terrestrial natural radiation sources (^{238}U, ^{232}Th, ^{40}K) and accounted for different thickness of concretes and room sizes. Due to the decay of ^{238}U and ^{232}Th, secular equilibrium was assumed.

2. BACKGROUND

2.1. Natural Background Radiation

Natural background radiation simply refers to radiation that occurs naturally without any man-made interference. It serves as the most abundant form of radiation exposure, contributing close to 80% of annual effective dose (Cinelli et al., 2019). Figure 2.1 shows the overall contribution of natural background radiation, medical radiation and other radiation sources for a total dose of 3mSv/y. Its origins could be traced back to the formation of the earth and its ubiquitous nature means it is present everywhere in the earth, commonly in rocks, oceans and building materials (Ramachandran et al., 2011). Human activities play a key role in most of the exposures effected by natural background radiation, this is seen through negligence in checking for its presence in building materials for construction. If overlooked, prolonged exposure can lead to lung cancer, hence, monitoring its presence is of high priority because understanding it, paves way to initiate appropriate radiation protection measures and protocols which in turn would help to reduce its detrimental effects (Al Khawlany et al., 2018).

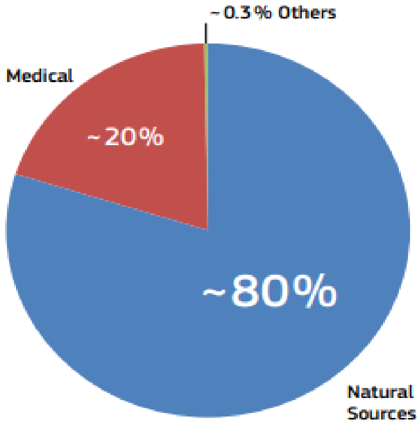

Figure 2.1. Chart showing the contribution of different sources of radiation (Cinelli et al., 2019)

2.1.1. Terrestrial Radiation and Radionuclides

The earth's total composition (ocean and land) has an abundance of natural radiation. These are primordial, evidently proven to have existed even before the formation of the solar system due to their significantly long half-lives. On average, these sources make up about 80% of the dose received yearly (Cinelli et al., 2019). However, the dose received vary significantly with geology and geography. The currently existing primordial natural radiation are those with half-lives comparable to the age of the universe, that is radionuclides with half-lives in the order of 10^{10} years, since they have not decayed much till present (Cinelli et al., 2019). These radionuclides are ^{238}U, ^{232}Th and ^{40}K. Other radionuclides also exist in nature but have minimal dose contributions that are practically undetectable and have no effect on humans.

Dose contribution to the human body which comes from terrestrial natural radiation results from gamma radiation released by the series of ^{238}U and ^{232}Th and from ^{40}K.

2.1.1.1. Uranium

Uranium (U) is a heavy actinide series element which has two main long-lived radionuclides, ^{238}U (half-life of 4.5 x 10^9 years and 99.274% of U total mass, more abundant) and ^{235}U (half-life of 7.0 x 10^8 years and 0.72% of total U mass, less abundant). Another form of uranium that exists in very small quantities and occurs as the third decay product of ^{238}U is ^{234}U (half-life of 2.4 x 10^5 years, 0.0056% of total U mass). All three forms of uranium radionuclides decay via alpha and beta emissions and when daughter nuclei are produced in their excited states, they emit gamma rays. The abundant ^{238}U undergoes radioactive decay through a series of 13 radionuclides before finally achieving stability in the form of ^{206}Pb. Gamma rays are produced from the decay products of ^{238}U.

2.1.1.2. Thorium

Thorium is a radioactive actinide series element with one primordial radionuclide ^{232}Th (half-life of 1.41 x 10^{10} years, 99.98% of total Thorium mass). It undergoes a long radioactive decay series with the stable lead isotope ^{208}Pb as its final decay product. Thorium also emits gamma rays from its decay products.

2.1.1.3. Potassium

Potassium is an alkali metal with three natural isotopes, two stable ones (^{39}K, 93.3% of total K mass and ^{41}K, 6.73% of total K mass), and the radioactive primordial ^{40}K (half-life of 1.248 x 10^9 years, 0.0117% of total K mass). ^{40}K decays into ^{40}Ca through beta particle emission or ^{40}Ar through gamma ray of energy 1.46MeV after electron capture.

In the human body, potassium is distributed uniformly and makes up approximately 0.18% of the body content. Of the total potassium content, the radioactive ^{40}K makes up 0.0117% and the rest are constituents of the stable ^{39}K and ^{41}K. About 10% of K decay in the body is released through gamma emission, the other 90% is absorbed in the tissues (Cinelli et al., 2019).

2.1.1.4. Gamma Energies of the Terrestrial Radionuclides

Detection of the terrestrial natural background radiation can be achieved by measuring the gamma energy it emits. ^{40}K decays to ^{40}Ar by gamma ray emission with energy of 1.46 MeV (emission probability of 10.67%) (Al-Khawlany, et al., 2018). The photopeak generated at the 1.46 MeV energy can be used to estimate the total amount of -^{40}K present in the sample being studied.

^{232}Th emits gamma ray in the range of 0.040 to 2.615 MeV (Orabi., 2016). However, ^{232}Th cannot emit gamma rays on its own, therefore, the gamma rays above are those of its decay products. ^{238}U also emits gamma rays from its decay products. Its energy range is 0.047 to 2.435 MeV (Orabi., 2016). Secular equilibrium is assumed for ^{238}U and ^{232}Th when investigating them due to the reason above (Cinelli et al., 2019).

2.2. Radon

Radon (^{222}Rn and ^{220}Rn) is considered the most significant contributor of natural background radiation to humans. Contributing about 52% of the total radiation mankind are exposed to yearly (Cinelli et al., 2019). It is a noble gas that is derived from the radioactive decay of ^{238}U and ^{232}Th. The ubiquitous nature of ^{238}U and ^{232}Th means that these are found in materials used for building structures. Figure 2.2 shows the contribution of radon among other sources.

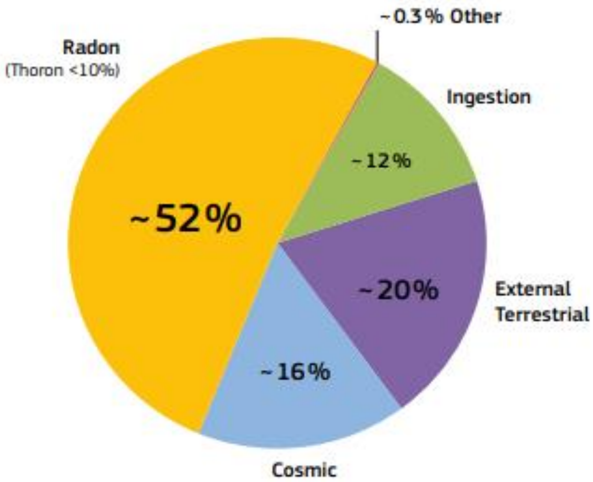

Figure 2.2. Chart showing the different contributions of the natural background radiations (Cinelli et al., 2019).

When the decay products of ^{222}Rn are inhaled, they are deposited along the walls of the bronchial tree with subsequent exposure to the lungs. The World Health Organization has stated that, after smoking, radon gas inhalation is the second leading cause of lung cancer, with its risk increasing by 16% in every 100 Bq/m^3 of exposure received (WHO, 2009). However, geological formations and its presence in building materials enhances the exposure to radon received by people in radon-rich areas (Al Jassim et al., 2018). The identification of radon as a common indoor air contaminant has led to a growing interest in studies related to its effects on human health (Veiga et al., 2003, Hendry et al., 2019, Sohrabi., 2013, WHO, 2009). The European Union has established a reference level of 300 Bq/m^3 as the level that buildings shouldn't exceed to keep exposure to these radon gas as low as possible.

2.2.1. Radon Source and Migration

Radon is present in all terrestrial materials such as the soil, rocks or building materials. This is because they are the decay products of radium (^{226}Ra and ^{224}Ra), which are decay products of the ubiquitous and primordial ^{238}U and ^{232}Th. Therefore, indoor radon exposure can result from the diffusion from soil. Indoor exposure to radon can also result from the advective transport of radon through the atmosphere in a process called radon exhalation (Cinelli et al., 2019).

Building materials mostly contain these primordial radionuclides and tend to release radon from their decay products. An example is blue concrete, a building material noted for contributing high indoor radon concentrations (Del Claro et al., 2017).

2.3. Effects of the natural background radiation on biological matter

Exposure to ionizing radiation has become part of the daily lifestyle of man. This is due to the vast number of sources that currently contribute to ionizing radiation, ranging from natural background radiation to artificial radiation contributed by industrial actions, medical practices, nuclear fallouts and many more (Ramachandran., 2011). When ionizing radiations like those from the natural background interact with tissues in a living organism in high quantities, it may cause detrimental effects to the cellular function or even modify the genetic code of the cells (Cinelli et al., 2019). This makes quantifying the dose received a vital factor, to which numerous varieties of quantities have been developed.

The most considered quantities are the absorbed and equivalent dose. Absorbed dose is the energy absorbed per unit mass and it is expressed in gray (Gy = J/kg). The equivalent dose links the absorbed dose and its biological effects. It is used to quantify the weighted sum of dose in tissue even in the presence of different contributions of dose by several radiations. It is calculated as the absorbed dose multiplied by the weighting factor of the radiation, a factor used in accounting for the relative biological effectiveness of various types of radiation. The equivalent dose is also expressed in J/kg or in Sievert (Sv).

Energy is not the only factor that influences the effects of ionizing radiation on biological matter, the type of radiation, DNA organization, how hydrated the DNA is and the way in which the energy is deposited are also crucial factors (Cinelli et al., 2019). This brings into contention the linear energy transfer (LET), defined as the mean energy deposited per unit path length in the medium of interaction, it is expressed in keV/μm. Factors such as the atomic number of the target and the particle's velocity influence the LET. Gamma rays have a low LET in comparison to other forms of radiation, that is why they are considered for medical purposes alongside X-rays but if threshold levels of exposure are exceeded, they become deleterious (Pouget et al., 2001).

2.3.1. Deterministic and stochastic effects

The effects of radiation on individuals can be characterised into two main ways, deterministic and stochastic effects. Deterministic effects are effects that do not occur below a certain threshold but above it, there is significant cell death which is enough to cause functional defects. Its severity is directly proportional to the dose, which means as the dose increases, the effect increases in response. The threshold of damage varies from person to person. They are acute and symptoms can range from nausea, headache, fever to skin and tissue burns. Symptoms can be noted as early as a dose of 0.5 to 1 Gy is received, however, mortality at this dose range is low and the symptoms can be managed. The mortality index increases and falls in the range of 5-95% for doses between 2-6 Gy received without treatment and a range of 5-50% when treatment is implemented. At 10 Gy and above, the mortality can reach 100% (Cinelli et al., 2019).

Stochastic effects are random effects and do not have a defined threshold. The probability of the effects is elevated with increased dose, but its severity does not rely on the dose. The effects often take time to manifest. Examples of these effects include cancer, hereditary defects, ocular defects, embryotoxic effects and many more. Hypothesis for stochastic effects have been difficult to evaluate due to the absence of information for the appearance of small stochastic dose curves. To ensure simplicity, it is assumed that it follows a linear non-threshold hypothesis.

2.4. The interaction of the gamma rays of natural background radiation

2.4.1. Photoelectric absorption

A tightly bound electron in an atom receives all the energy of a gamma ray when they collide. The electron uses some of the energy to overcome the binding energy and free itself from the atom. Figure 2.3 below shows an incident gamma photon interacting with an atom and the resulting escape of a photoelectron (Nelson et al., 1991).

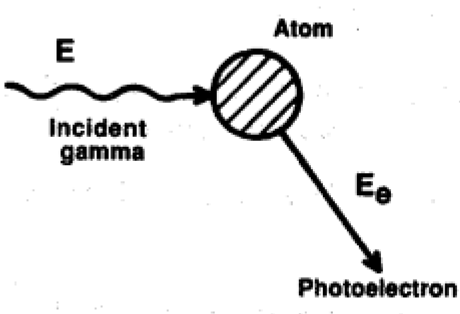

Figure 2.3. An incident gamma photon interaction with an atom and the resulting release of a photoelectron (Nelson et al., 1991).

The photoelectron that is ejected from the atom has an energy (E_e) which results from the difference between the gamma ray energy (E) and the electron binding energy (E_B) as seen below,

$E_e = E - E_B$ **equation 2.1**

The rest of the energy is used to propel the electron in the form of kinetic energy out of the atom. There is a small amount of recoil energy left in the atom for the conservation of momentum. The importance of photoelectric absorption in gamma ray detection lies with the full energy of the gamma ray being resolved into a full-energy peak. Information obtained from the full-energy peak can be used to identify unknown sources (McAlister, 2012). The interaction is very significant for heavy atoms like uranium, low-energy gamma rays, x-rays and bremsstrahlung. A decrease in the gamma energy also leads to a rapid increase in the photoelectric absorption.

2.4.2. Compton Scattering

In this interaction process, a gamma ray interacts with a loosely bound electron in the outer shell and transfers a part of its energy to the electron. The electron being lose means that it does not need much energy to overcome the binding energy of the atom. The energy transferred to it by the gamma ray is the energy it is ejected with. Figure 2.4 shows this interaction. The gamma ray uses the rest of the energy left to scatter in a different direction.

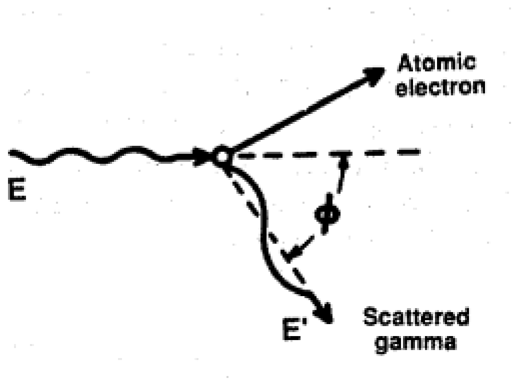

Figure 2.4. Interaction of a gamma ray with a loosely bound electron and the resulting Compton scatter of the electron and the gamma ray after energy loss (Nelson et al., 1991).

The energy of the scattered electron is expressed below,

$E_e = E - E'$ **equation 2.2**

Where E_e is the energy of the scattered electron, E is the energy of the incident gamma ray and E' is the scattered gamma ray's energy. The direction that the electron and scattered gamma ray takes after the interaction is dependent on the energy transferred to the electron during the interaction. Expressed below is the mathematical representation of the energy of the scattered gamma ray (E') as a function of direction.

$$E' = \frac{E}{1+\frac{E}{m_0c^2}(1-\cos\theta)}$$ **equation 2.3**

Where E is the energy of the incident photon, θ is the angle between the incident and the scattered ray and m_oc^2 is the rest mass energy of the electron (511 keV). If the angle is 180° (a result of head on collision), the energy is minimum because the electron moves in the direction of the gamma ray leading to backscatter. The energy of the scattered gamma ray then becomes,

$$E' = \frac{E}{1+\frac{2E}{m_0c^2}}$$ **equation 2.4**

If the scattering angle is very small ($\theta \approx 0^0$), the energy of the scattered gamma ray is marginally less than the energy of the incident gamma ray leading to less energy for the scattered electron. The scattered electron's energy ranges from zero to a maximum energy given by equation 2.4 above**.**

Probability of Compton scattering depends on the electron density, which is proportional to Z/A, where Z is the atomic number and A is the atomic mass. Due to the sole involvement of the least bound electrons, the influence from the nucleus on the process is negligible.

A typical gamma spectrum showing photoelectric absorption and Compton scattering is observed in figure 2.5 below.

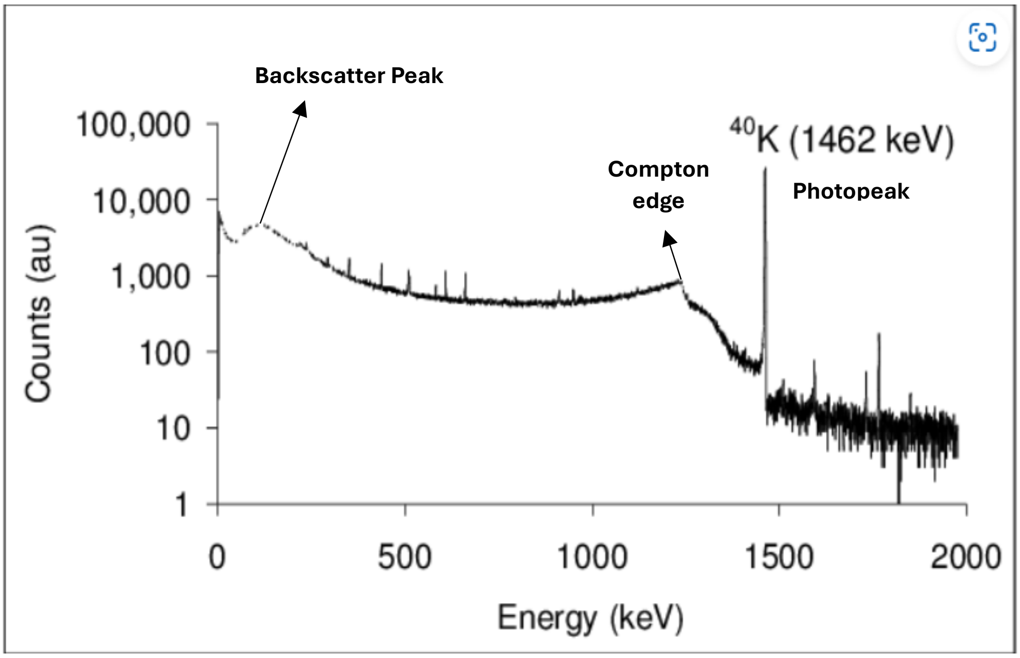

Figure 2.5. A gamma spectrum of ^{40}K measured as a result of photoelectric and Compton interactions (Espinosa et al., 2009).

The main photopeak at 1462 keV occurs because of the gamma ray losing all of its energy by photoelectric absorption in the detector. The events that precede the photopeak is related to the Compton scattering that happens alongside the photoelectric absorption. The part labelled as the Compton edge is the maximum energy an electron could receive by a 1462 keV ^{40}K gamma ray in a single Compton scattering process. The backscatter peak seen closer to the 256 keV mark is formed from the ≈180° scattering that occurs. The sum of the Compton edge and the backscatter peak is the same as the energy of the incident gamma ray. The backscatter peak happens when the scattered gamma ray deposits its energy into the detector while the Compton edge happens when the scattered electron deposits its energy into the detector.

2.4.3. Pair Production

In pair production, a gamma ray with an energy greater than or equal to 1.022 MeV can cause the ejection of a positron-electron pair when it gets close to its strong electromagnetic field.

The gamma ray disappears after the interaction. The threshold energy of 1.022 MeV is so because it is the minimum energy required to create the positron-electron pair (Nelson et al.,1991). Any extra energy from higher energies than the threshold is converted into kinetic energy for the pair. Pair production is illustrated in figure 2.6 below.

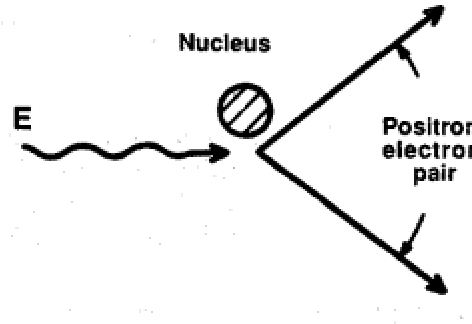

Figure 2.6. A diagram showing the interaction of pair production, and the positron-electron pair produced.

2.5. Attenuation

Attenuation refers to the reduction in the intensity of the radiation beam as it passes through a material. This is of great importance when radiation protection is being considered indoors to prevent exposure to natural background radiation. Mainly because the more the material can attenuate, the less radiation beam is allowed to pass through. Gamma radiation attenuation is calculated by the equation below,

$I = I_o e^{-\mu x}$ **equation 2.5**

Where I is the intensity of the gamma ray after shielding, I_o is the incident intensity, μ is the attenuation coefficient and x is the thickness of the material. However, the attenuation of materials is dependent on the mass of the material so density (ρ) is also considered. Therefore, equation 2.5 above can be modified to account for mass attenuation as,

$$\frac{\mu}{\rho} = \frac{-\ln\left(\frac{I}{I_0}\right)}{x} ,$$ **equation 2.6**

Where, $\frac{\mu}{\rho}$ is the mass attenuation coefficient.

Mass attenuation is high for low energies and low for high energies. This is because photons of high energy contribute more energy to the target and more penetrates the material, low photons on the other hand contribute less and more are attenuated, hence the high attenuation. higher for high atomic number materials such as lead, bismuth and tungsten at low gamma energies (<500 keV) than low atomic number materials such as iron and aluminium. This means that the high atomic number materials can effectively shield low energy gamma rays than low atomic number elements. For energy range of 800-1400 keV, there are similarities between the mass attenuation coefficient of the materials irrespective of atomic number. Attenuation by photoelectric effect is dominant for high atomic number materials at low energies (<500 keV). Attenuation by Compton scattering is dominant in low atomic number materials at energies higher than 500 keV (McAlister, 2012). Using the energy range of 800-1400 keV as a reference, shielding by an equal mass of lead and iron will decrease the gamma energy by equal amounts respectively, but the dominance of Compton scattering in iron means that it would not be able to reduce more gamma dose than the lead at the same mass. Figure 2.7 below shows a graph of total mass attenuation coefficient of some shielding materials against gamma energies (McAlister, 2012).

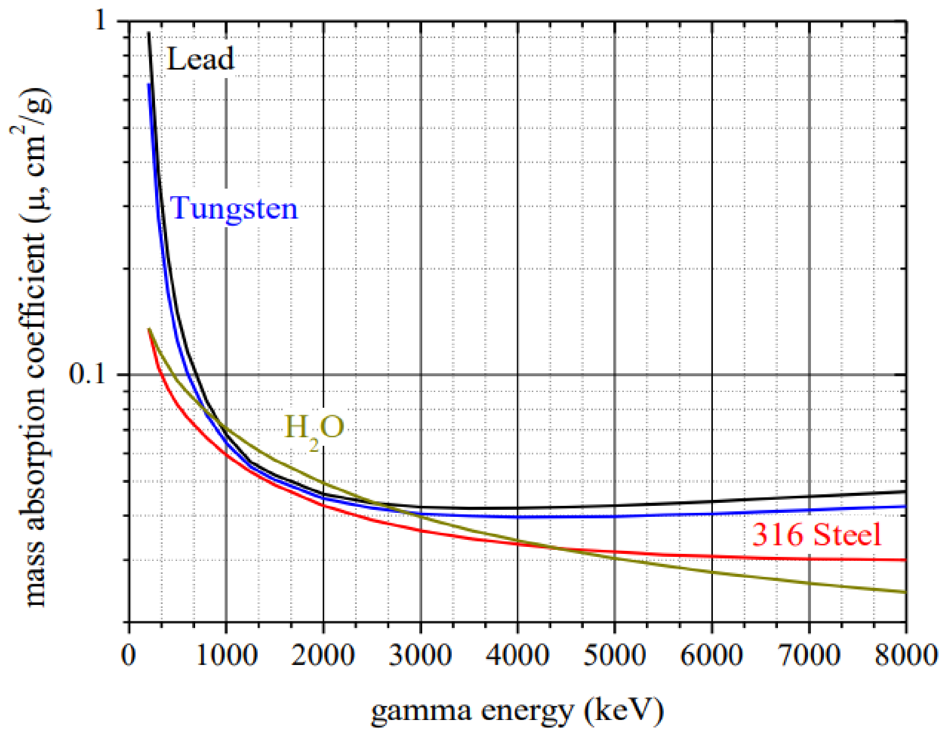

Figure 2.7. Mass attenuation coefficient against gamma energy for common shielding materials (McAlister., 2012).

3. LITERATURE REVIEW

3.1. Natural Radiation in Building Materials

Building materials contribute significant levels of natural background radiation to individuals due to the presence of radionuclides in them. The need to investigate these radionuclides arises from humans spending 80% of their time indoors (Mehdizadeh et al., 2011). In the UK, for instance, the average person spends 8% of their time outdoors (Al-Khawlany, et al., 2018). This means that exposure received indoors from the gamma radiation of these radionuclides are quite significant and if left unchecked, can lead to radiation risks. These radionuclides are mostly of terrestrial sources like soil and rocks, which are widely used in manufacturing building materials. Radon gas also contributes to indoor exposure through emission from the soil underneath the building.

The natural radioactivity concentration in building materials has been estimated in many countries to ensure that the values conform to the normal levels that have been outlined (Beretka et al., 1985, Othman et al., 1994, Alam et al., 2001, Khan et al., 2001, Faheem et al., 2008, Al-Khawlany, et al., 2018, Ridha et al., 2013).

As much as the indoor exposure is significant, certain factors such as the soil type, building structure and geometry, and individual lifestyle also influences the dose received (Appleton., 2014).

3.2. Radiation Risk Associated with Natural Background Radiation exposure and study designs for better analysis of the risk

Radon, as a natural radiation, contributes the highest exposure to mankind and has been categorised as the second leading cause of lung cancer (WHO., 2009), hence, it has become imperative to investigate the level of risk it poses to better alleviate them, especially in high natural background radiation areas (Sohrabi, 2013, Hendry et al., 2009). Radon and its association to lung cancer has long been demonstrated in a miner cohort in the Biological Effects of Ionizing Radiation of the National Academy of Science (BEIR VI., 1999). This motivated the research of indoor radon to see if there is a risk association like the one found in the miners.

Veiga et al., (2003) investigated the radon exposure received by inhabitants in the rural and urban areas of Poços de Caldas in Brazil, a high background natural radiation area. Both external radiation and indoor radon concentrations were assessed to evaluate their relationship to lung cancer risks among the population. It was noted that, the rural areas had a high background radiation level than the urban areas. The indoor radon concentration levels for the urban areas ranged from 12-920 Bqm^{-3} with a mean of 61 Bqm^{-3} and the rural areas had concentrations ranging from 27-1024 Bqm^{-3} with a mean of 220 Bqm^{-3}. It was also inferred that ^{222}Rn contributes the most effective dose to the rural population (3.69 mSv/y), while gamma irradiation contributed the second most effective dose of 1.32 mSv/y, ^{220}Rn contributed the least (0.7 mSv/y). Veiga et al., (2003) attributes the high values of indoor radon concentration in the rural areas to the geological formation, building materials and the ventilation features of some houses.

A European pooling study that incorporated certain study size limits, a minimum of 150 case and control sample for lung cancer studies per each European country that participated also investigated the relationship between radon and lung cancer (Darby et al. 2005, 2006). The criteria also incorporated the smoking data of the participants and ensured a least of 15-year history of radon measurement data on the homes of the participants were available. As a result, a good total of 7000 cases and 14000 controls were entered into the study from 13 different European countries. This pooled analysis helped to identify a positive association between radon and lung cancer. Results showed that the lung cancer risks increased by 8% per 100 Bqm^{-3} increase in measured radon concentration (Darby et al., 2005, 2006, WHO., 2009). Furthermore, it was observed that the study demonstrated a linear non threshold dose response. When a threshold of 150 Bqm^{-3} was set, the results did not necessarily identify proof that the effects happened above that threshold because there were other effects that happened below it, however it was noted that, the risk of lung cancer increased by 20% in participants that were exposed to radon concentrations of 100-199 Bqm^{-3} in comparison to individuals exposed to values that were less than 100 Bqm^{-3}.

Similar pooling studies have been conducted in North America and China, with the risk of lung cancer increasing by 11% per 100 Bqm^{-3} and 13% per 100 Bqm^{-3} respectively (Krewski et al. 2005, 2006, Lubin et al., 2004). All three studies demonstrated a linear non threshold dose response

Barros-Dios et al., (2002) studied the relationship between smoking, radon and lung cancer in a cohort of smokers and non-smokers. The authors noted that for non-smokers, the chance of developing lung cancer by the age of 75 years from exposure to radon concentrations of 0, 100 and 400 Bq/m^3 were about 0.4%, 0.5% and 0.7%, respectively. Smokers, on the other hand, had a higher chance of developing lung cancer (25 times greater) at the same concentrations at 75 years, about 10%, 12% and 16% respectively.

3.3. The Need for Radiation Protection, Agencies involved in its Implementation and Approaches to achieve it.

The confirmation of a positive association between radon exposure and lung cancer gives rise to the need to protect inhabitants in high background natural radiation areas. As such, agencies have been formed with the aim of putting measures in place to protect these inhabitants. The International Commission on Radiation Protection (ICRP) is one of such agencies charged with safeguarding the general population against ionizing radiation effects. The commission recommends a reference exposure level of 300 Bqm^{-3} to radon and its progeny based on some of the new risk findings, a decrease from its initial recommendation of 600 Bqm^{-3} (Clement et al., 2010, Lecomte et al., 2014). This value is used as a baseline for national authorities in the setup of current and future radon reference levels, ensuring that radon concentrations are lowered. The World Health Organization has a dedicated indoor radon handbook which is constantly updated with new trends to aid in accounting for radiation protection (WHO, 2009). Radon legislations have also been slated out by the International Atomic Energy Agency (IAEA), they achieved this by receiving data from 15 European Union (EU) states and 17 non-EU countries on the radon levels present in their environments and the threshold values enforced to combat its effects (Aakerblom., 1999). Many other organizations also exist both locally and nationally that are charged with ensuring the safety of individuals by implementing and enforcing radiation protection regulations.

Apart from radiation protection agencies, other bodies of work have recommended approaches for ensuring radiation protection. Sohrabi (2013) proposes two of such approaches in the study of high background natural radiation areas. The first approach involves an extensive remodelling of buildings and amenities through diverse planning and intentional renovations. The author further weighs the advantage of individuals keeping their settlements against a voluminous demerits like high renovation costs, the impossibility of reducing every radioactivity in the area, and many others which leads the author to a second approach. The second approach involves the mass migration of inhabitants to a normal radiation area and the conversion of the high background natural radiation area into an environmental radioactivity park, Sohrabi (2013) suggests that this method is cost-effective but in essence, this approach seems more costly than the first approach, this is because cost would be factored into building new homes for these new inhabitants, compensation would have to be offered to the migrants to help them start a new life, the conversion of the area into a radioactivity park will also factor some kind of high cost, it can be very difficult to even convince certain inhabitants to abandon their old homes.

A more practical approach is proposed in which soil depressurization is performed in a model house built in a high radon area in Spain. Vazquez et al., (2011) investigated the soil depressurization technique in the model house and observed a radon level concentration reduction in the range of 91-99% except in the passive ventilation and outside sumps where the levels were reduced by 53-55%. Another study done in the same house investigated radon reduction by soil depressurization which considered both active and passive ventilation, results showed that radon reduction was achieved in excess of 85%, further postulating that a fan power of 20W is sufficient to elicit this reduction in houses with similar aggregate layer and soil permeability (Fuente et al., 2019). Geant4 simulations can also be capitalized to investigate and plan the necessary measures that can reduce the radiation in the area.

3.4. The role of Geant4 Monte-Carlo Simulations in the investigation of Natural Radiation.

Geant4 is a software toolkit used to simulate the passage of particles through matter (Allison et al., 2016). Its role in investigating natural radiation has seen an increase due to its ease of use, coupled with its accurate and realistic results. Many studies have explored this in investigating different natural radiation, some of those studies relevant to this work are discussed below.

Li et al., (2016) proposed a novel Monte-Carlo simulation model on natural background by simulating the decay of ^{238}U, ^{232}Th and ^{40}K in soil. A spectrum of energies was obtained using the Monte-Carlo simulation ranging from 73.9keV to 2615keV. This was then compared to a measured laboratory spectrum of the same range and the values agreed with the laboratory reference, especially, energies above 250keV. To verify the usability of the model, it was used in the study of the measurements of an anticoincidence detector for a radioactive point source in natural background. It was inferred that the anticoincidence measurements reduced the natural background and improved the detectability of the activity compared with normal measurements. Hence, establishing the model’s ability to detect low radioactivity.

Hung et al., (2017) investigated cosmic ray spectrometry using Geant4. They measured natural background gamma energies within the range of 0.04 to 50MeV. The measured values were compared to a spectrum of laboratory reference and there was good agreement between the two in the range of 3-50 MeV. The contribution of cosmic -ray components such as muons, neutrons, protons, electrons, positrons and photons were also investigated. The results showed that muons dominated, contributing about 86% amongst the components that were observed from the cosmic rays. The contribution of the other components was noted below the 3MeV energy region.

Orabi (2016) investigated factors that affected indoor gamma radiation dose rate using a Monte-Carlo Simulation software. Factors such as thickness of the walls and the density of the building materials increased as the dose rate increased until a point of saturation due to the walls self-absorbing the energies they emitted. Different room dimensions were also investigated, and it was noted that, as the room size increased, the dose rate that was detected also increased.

4. SIMULATION METHODS

A tool for Particle Simulation (TOPAS) code was developed to investigate the dose received by an individual from the gamma energies of the natural background radiation in a building made up of concrete walls (Faddegon et al., 2020). The code for the simulations is in the appendix.

4.1. Natural radionuclides and their energies used in the simulation

The natural radionuclides considered for this study were ^{40}K, ^{238}U and ^{232}Th. ^{40}K has one gamma energy at 1.46 MeV. It was assumed that the decay products of Uranium and Thorium are in secular equilibrium with ^{238}U and ^{232}Th respectively. Exhalation of ^{222}Rn (for ^{238}U) and ^{220}Rn (for ^{232}Th) were not considered. The photon energies in both decay series with the most contribution to dose were considered for this study. Table 4.2 shows these photon energies and their intensities.

Nuclide	Energy (keV)	Intensity (%)
U series		
^{214}Pb	351.932	37.6
^{214}Bi	609.312	46.1
Th series		
^{212}Pb	238.632	43.3
^{208}Tl	583.191	30.4
^{228}Ac	911.204	25.8
^{40}K	1460.83	11

Table 4.1. Radionuclides of natural background radiation, their energies and intensities (Clouvas et al., 2000). (Pb – lead, Bi – Bismuth, Tl – Titanium, Ac – Actinium).

4.2. Geometry of the original test room (model house) and the individual

A model house was defined in the code for the purpose of indoor investigations in this study. The house was made up of a room of dimensions, 5x4x2.8 m^3, and 4 wall thickness of 20 cm made up of concrete (Orabi, 2016). The floor and ceiling of the room were also made up of concrete, with thickness of 20 cm, same as that of the walls. There were no windows and doors for the room. A cylinder of height 1.8 m and radius 10 cm, made up of water was placed in the room. This was representative of a human being since water makes up the most abundant compound in human beings (Kleiner, 1999). Hence, for clarity, further reference to the cylinder will be as an "individual". Figure 4.1 shows a graphical view of the model house, and the individual generated from the code.

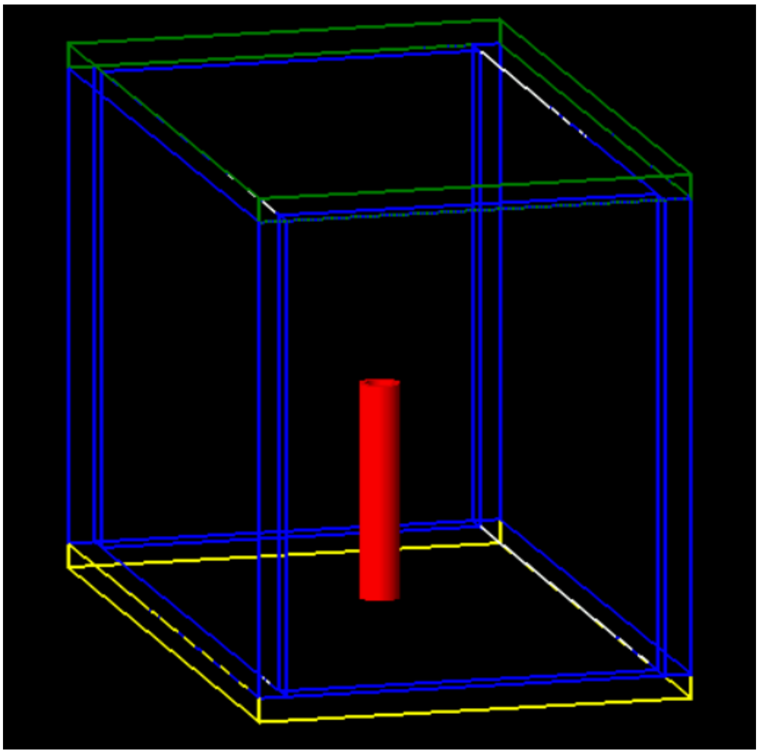

Figure 4.1. A room of dimensions 5 X 2.8 X 4 m^3, surrounded by walls of concrete, a ceiling (colour-coded as green) and a floor (colour-coded as yellow), all with thickness of 20cm. A red cylinder of height 1.8 m is in the centre of the room.

Dose absorbed by the individual in the simulation was recorded for analysis. The position of the individual in the room was close to the floor to ensure realism in the investigation, as if the individual was standing in the room. For most of the simulation, the individual was at the centre of the room which is 1.5 m from the big wall (5 x 4 m^2) and 2.6 m from the small walls (2.8 x 4 m^2).

4.3. Defining the walls, ceiling and floor as the source of gamma rays.

The simulations were based on the gamma energies emitted from building materials by terrestrial radionuclides. Hence the walls, ceiling and floor of the house were defined to be the source of radiation in the code. The gamma radiation emitted were from their concrete composition. The radiation emitted were isotropic and omnidirectional. This is shown in figure 4.2 below.

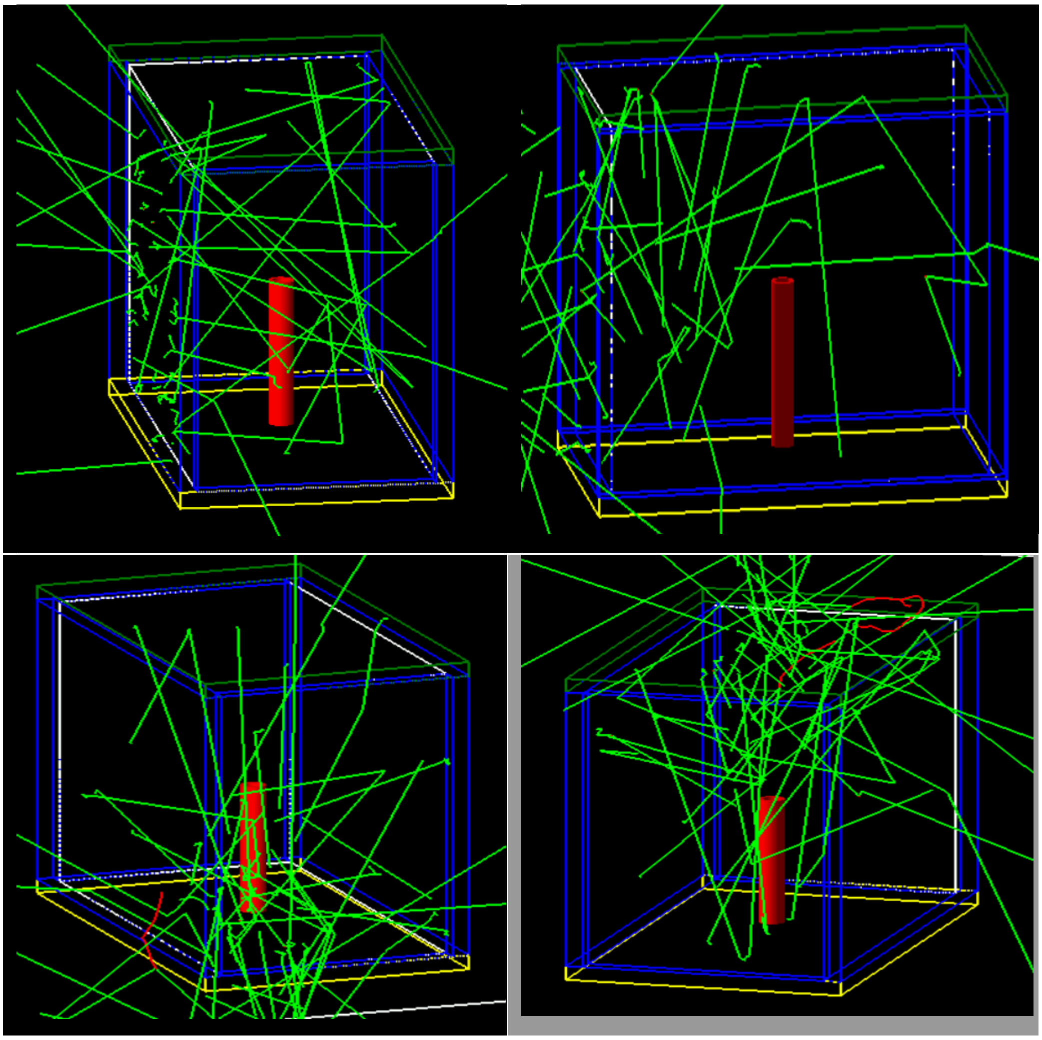

Figure 4.2. Radiation emission from the different parts of the room. Upper left: Emission from the big wall, upper right: Emission from the small wall, lower left: Emission from floor, lower right: Emission from the ceiling.

The composition of the concrete used in the simulation as defined by GEANT4 is listed in table 4.2 below. The density of the concrete is 2.3 g/cm^3.

Element	H	C	O	Na	Mg	Al	Si	K	Ca	Fe
Composition	0.010	0.001	0.529	0.016	0.002	0.034	0.34	0.013	0.044	0.014

Table 4.2. Elements and their compositions in the make-up of Concrete (Allison et al., 2016).

It should be noted that TOPAS allows beam definition in one geometry at a time. Due to this, examining the dose contribution from each geometry will require that the radiation is defined in that geometry. That is why figure 4.2 shows different position of the emissions instead of a full contribution from the whole house. The usefulness of this feature is that it allows the individual quantification of dose received from multiple geometries (walls, ceiling, floor) which is great from a statistical point of view. This is because although the compositions of the geometries are practically similar, there are bound to be nuances that might lead to variations in the dose received from each geometry and knowing that also helps in adjusting the simulation or influences the inference made from it.

4.4. Test Case of the building material

The building material (concrete) was subjected to a variety of tests to ensure that results obtained from the simulation were in line with the physics that were expected. The tests involved the simulation and calculation of the mass attenuation of concrete, the variation of the attenuation with different thicknesses and gamma spectroscopy. The gamma energy used for this test case was the 1.46 MeV of ^{40}K. This was used specifically because its photopeak in spectroscopy is just one, at the 1.46 MeV, and hence resolving it is easier than the numerous emissions of the other decay series. It also provided sufficient energy for the attenuation simulations.

4.4.1. Attenuation Simulation set-up and simulations for the test case

A simple set-up of a 20 cm thick concrete wall placed between a spherical radiation point source and germanium detector (50 x 50 x 10 cm^3) was defined in a separate simulation code (See Appendix). The energy from the radiation source was from the radionuclides listed in table 4.1. The simulation was run for 100,000 events and some of the energies were attenuated by the wall while the remainder were recorded by the detector. Figure 4.3 illustrates the set-up.

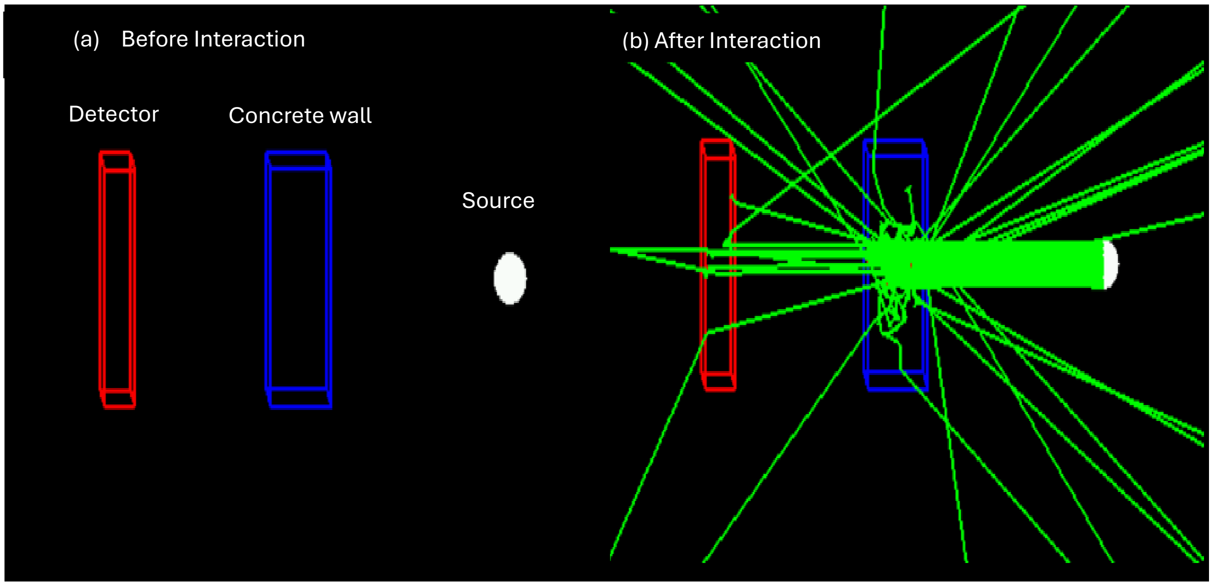

Figure 4.3. Test case set-up showing attenuation of radiation by the concrete wall

From this simulation set-up, the mass attenuation coefficient of concrete was calculated using the equation given below.

$$\frac{\mu}{\rho} = \frac{-\ ln\left(\frac{I}{N(I_0)}\right)}{x} \qquad \textbf{equation 4.1}$$

Where N is the number of events (histories) that were simulated, and the other expressions are the same as those in equation 2.6. The number of histories (N) or events is defined as the set of information about the particles, the behaviour of these particles and their subsequent secondary particles produced after interaction or energy deposit (Allison et al., 2016). This is pivotal in TOPAS because it provides the necessary information about the type of particle being investigated. It can also be explored to investigate different parameters of the particle like increased dose, high interactions and many other advanced interactions. For better statistical coherence, a higher number of histories is required. In this study, 100,000 events (histories) were used as the baseline because it was sufficient to provide the expected results in the simulation.

The values obtained after the simulation were compared to standard mass attenuation coefficient values. These standard values are already measured and a database of them for different energies and compositions is provided by the National Institute of Standards and technology in a program called XCOM (Berger, 2010).

Different thicknesses of concrete were also simulated with the same set-up to investigate how thickness varied with the attenuation.

To obtain the gamma spectrum, 200 bins were introduced into the detector and the energy recorded in each bin was counted. The simulations were defined to count 10 keV per bin and was run for 100,000 events for better statistics. A graph of counts against energy was then plotted to represent this.

4.5. Simulations in the test room and different room sizes

Dose simulations were performed using the gamma energies in Table 4.2. The measurements accounted for dose received at the centre of the room and at corners (near walls). Figure 4.4 illustrates the position of the individual in the simulation. The source of the beam was the walls, ceiling and floor. TOPAS allows beam definition for one geometry at a time, therefore, the dose received was simulated for all the 6 geometries (4 walls, floor and ceiling) as illustrated in figure 4.2.

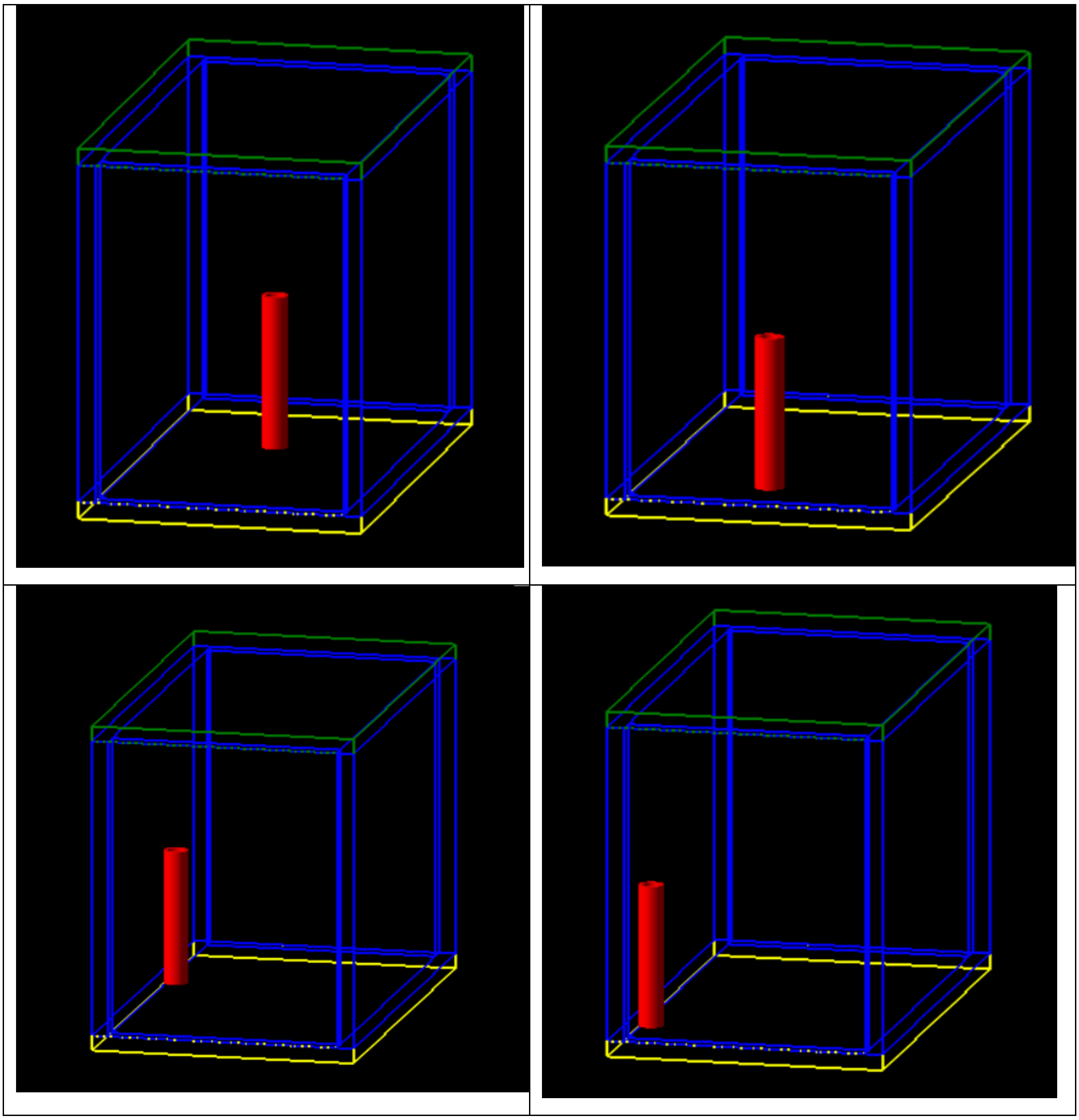

Figure 4.4. Different positions of the individual inside the building for indoor dose measurements.

The thickness of the big wall was also increased while accounting for increasing composition and simulations were performed with the individual at the centre of the room. This was done to investigate the effects of high composition of the natural radionuclides, which results from increasing the building constituents from its increasing thickness, on dose received by the individual. The simulations were done for the high (1.46 MeV) and low energies (0.24 MeV)

Lastly, different room sizes were defined and simulations from the terrestrial radionuclides in them were investigated. This was done to investigate how room sizes impacted the dose received. For simplicity, the thickness of the room sizes remained unchanged and was defined as 20cm, same as the thickness for the original model house.

5. RESULTS AND DISCUSSION

5.1. Test-Case Simulation

5.1.1. Attenuation Coefficient

The test case set up shown in figure 4.3 was simulated for 100,000 events for each of the gamma energies and the energy deposited into the detector after attenuation was recorded. Table 5.1 shows the calculated mass attenuation coefficient from the simulations and their corresponding measured standard values, obtained from using XCOM (Berger, 2010) to investigate the attenuation coefficient from the composition of concrete (See Table 4.1).

Energy (MeV)	**Radionuclide**	**Mass attenuation Coefficient (cm^2/g) (From simulation)**	**Mass attenuation Coefficient (cm^2/g) (From XCOM)**
0.35	^{214}Pb	0.0815 ± 0.082	0.1008
0.61	^{214}Bi	0.0656 ± 0.013	0.0781
0.24	^{212}Pb	0.0935 ± 0.024	0.1208
0.58	^{208}Tl	0.0668 ± 0.013	0.0797
0.91	^{228}Ac	0.0548 ± 0.010	0.0648
1.46	^{40}K	0.0430 ± 0.074	0.0529

Table 5.1. Calculated mass attenuation coefficient of concrete from the gamma energies of the radionuclides, obtained from simulation and compared to standard mass attenuation coefficient.

From the table, it could be observed that the data calculated from the simulation agreed with the standard values. The simulated values were not as precise as the values obtained from XCOM but the uncertainty range accounted for the agreement. Further increase in the number of events to 1,000,000 did not have any effect on the attenuation coefficient. The mass attenuation co-efficient values for low energy gamma rays like 0.24 MeV of ^{212}Pb and 0.35 MeV of ^{214}Pb were higher in comparison to that of high energy gamma rays like 0.91 Mev of ^{228}Ac and 1.46 MeV of ^{40}K. This is representative of photoelectric absorption dominating at lower energies which leads to a higher attenuation of the radiation as it passes through a material like concrete and Compton scattering dominating at high energies (McAlister, 2012). This confirmed that the right physics was being simulated in the code and results obtained were in line with theory.

5.1.2. Gamma Energy Spectrum of ^{40}K

A second test case involved exploring whether the photopeak of the ^{40}K radionuclide could be resolved from the set-up. The 1.46 MeV gamma energy of ^{40}K was used specifically because its photopeak in spectroscopy is just one, at 1.46 MeV, hence, it was easier to test it in the simulation without consideration for the presence of secondary photopeaks. The test ensured that the actual energy defined was the one recorded. Figure 5.1 shows the results from the simulation.

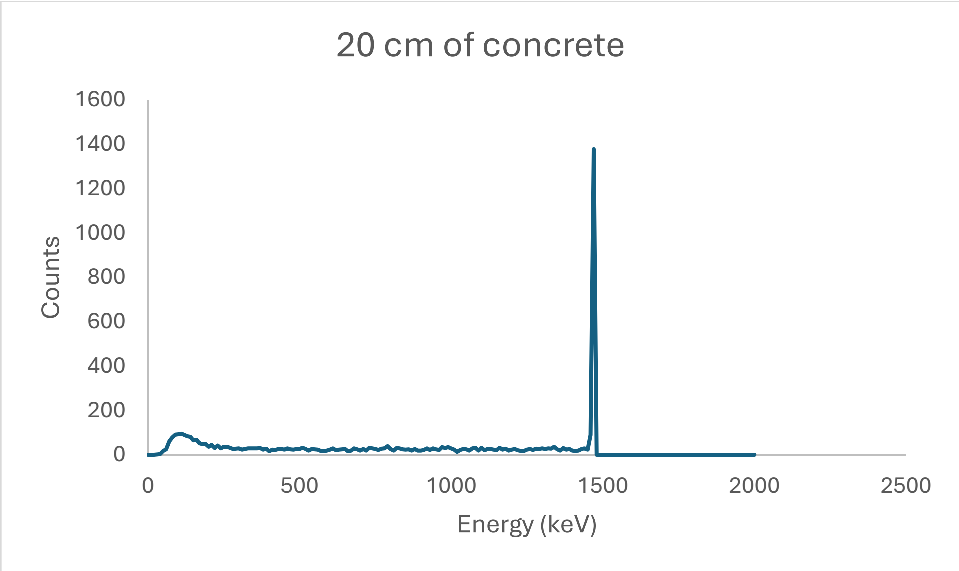

Figure 5.1. Gamma Energy Spectrum of ^{40}K obtained from TOPAS

It should be noted that TOPAS is not comprehensively built to fully resolve spectrums, therefore it was a bit difficult to showcase Compton background properly. However, it capably resolved the photopeak at the 1.46 MeV energy mark which also verified that the correct energy was being detected, hence verifying the feasibility of the simulation for complex build.

5.1.3. Attenuation and Thickness

The final test case to verify the concrete for the simulations was performed to investigate how increasing thickness of the concrete affected attenuation. This was also done for the 1.46 MeV energy of ^{40}K. Figure 5.2 illustrates the results obtained for this. The result demonstrated that higher thickness attenuates more of the energy and is consistent with an experimentally measured study that utilizes a similar set-up as figure 4.3 (Adedoyin et al., 2016). This also verified that the concrete automatically reduces the intensity of the gamma ray with increasing thickness.

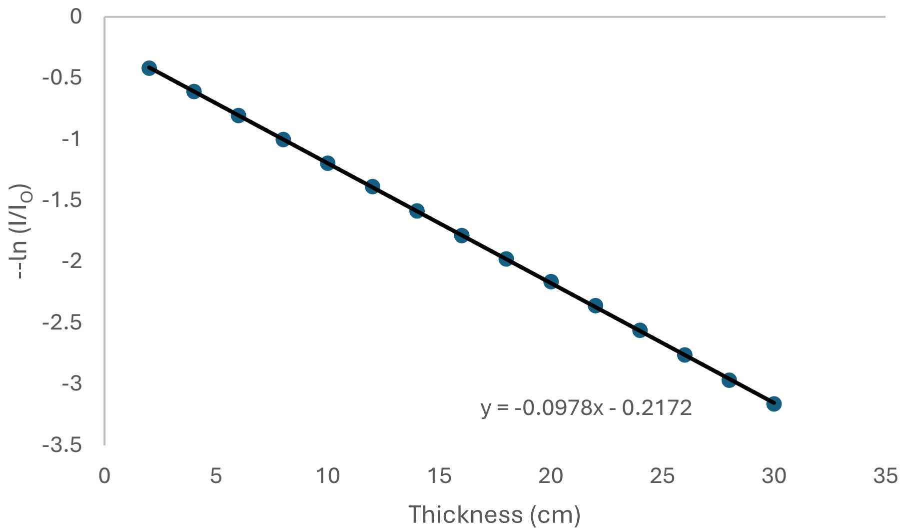

Figure 5.2. Attenuation of gamma ray energies in different thickness of concrete for the gamma energy 1.46 MeV of ^{40}K.

5.2. Simulations to validate the gamma radiation emitted by the walls, ceiling and floor and investigate dose received near the walls

5.2.1. Validation of gamma radiation from walls, ceiling and floor

To validate the emission of gamma radiation from the building components (walls, ceiling and floors), gamma energy in different thickness of the big wall (5 x 4 m^2) in the model house was simulated while the Individual stayed at the centre of the room. The gamma energy used for this validation was 1.46 MeV of ^{40}K. The thickness chosen for the simulations ranged from 2 cm to 40 cm and was increased evenly (that is, 2 cm, 4 cm, 6 cm, ……, 40 cm). TOPAS is designed in such a way that it prioritizes the shielding properties of beam geometries with increasing thickness (As seen in section 5.1.3). This means that increasing the thickness alone causes the concrete wall to attenuate the radiation without accounting for additional composition. To resolve this, the number of events for the thicknesses were increased by a factor of 10,000 for each thickness to exhibit an increase in the composition of the walls and hence, an increase in the radiation emitted (2 cm was simulated for 10,000, 4 cm for 20,000, ……, 40 cm for 200,000). The number of events in this case acted as the quantity of radioactivity present in the wall, which practically increases with increasing thickness because more building materials are theoretically added which means that more natural background radiation is added to the wall. Figure 5.3 shows the dose to the individual for the 1.46 MeV energy of ^{40}K.

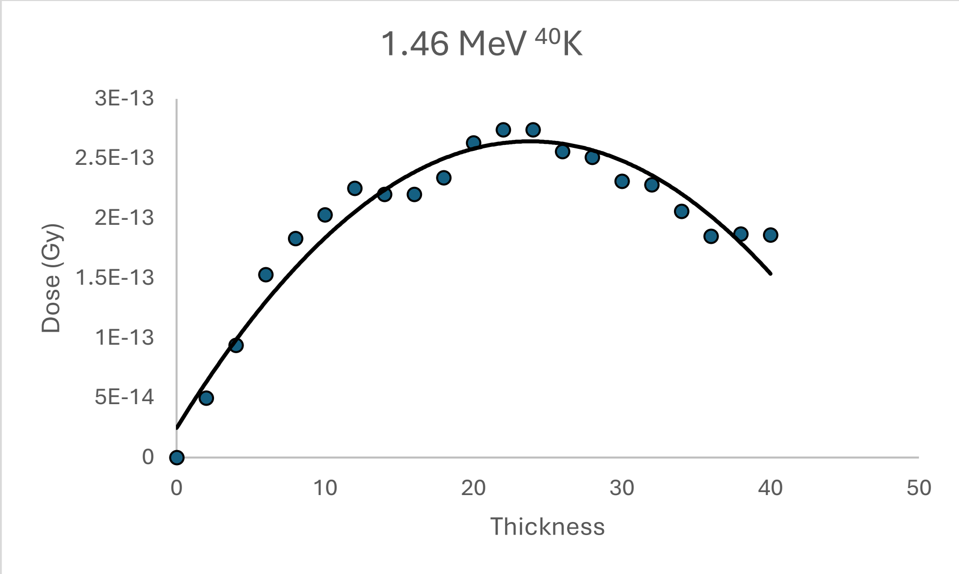

Figure 5.3. Dose as a function of increasing thickness for the ^{40}K radionuclides. Number of events were increased for the thickness to demonstrate increased composition.

The dose increased with increasing thickness until it reached the thickness of 12 cm. The dose started to saturate above the 12 cm thickness, fluctuating between the 2.5 x 10^{-13} and 1.5 x 10^{-13} Gy range. This occurred because the simulation demonstrated self-absorption, a phenomenon that originates from the wall acting as both a source and a shield to its own radiation (Tsutsumi et al., 2001). This means that at the thickness of 12 cm and above, the wall attenuated some of its gamma ray energy, although the composition increased with thickness. Others that were emitted were of low energy and could not contribute any dose to the individual. A similar simulation was done for the 0.24 MeV energy of ^{212}Pb (the lowest energy among the radionuclides) and the results are presented in figure 5.4. For the low energy, the dose increased initially like that of the high energy, however, saturation started earlier at the 6 cm thickness and occurred until the 18 cm thickness. Afterwards, a rapid decline in dose was observed. Compared to the high energy (1.46 MeV), the wall attenuated more of the gamma energy released by the low energy (0.24 MeV), hence, it was not able to contribute significant dose to the individual like the high energy. Saturation also lasted longer for the high energy than the lower energy, this is because although the composition increased, the lower energies were not strong enough to be emitted from the increasing thickness of the wall. This also accounted for the rapid decline in the low energy than the high energy.

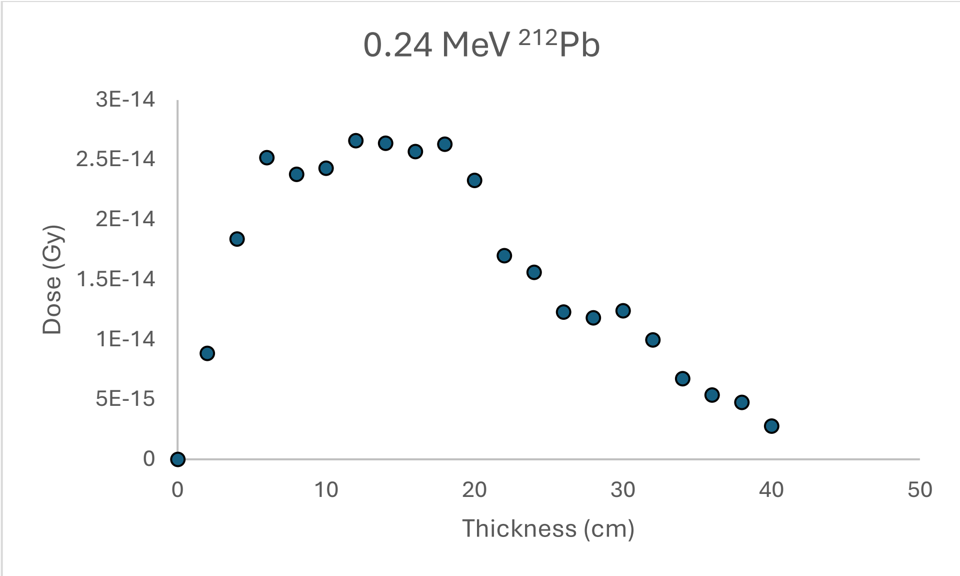

Figure 5.4. Dose as a function of increasing thickness for the ^{212}Pb radionuclides. Number of events were varied for the thickness to demonstrate increased composition.

5.2.2. Near wall simulation

Figure 5.5 provides details on some of the measured doses near the walls of the room, specifically at the corners, for the gamma energy of ^{40}K. Similar simulations for the gamma energies of ^{214}Pb of ^{238}U and ^{208}Tl of ^{232}Th were also done and are presented in the appendix.

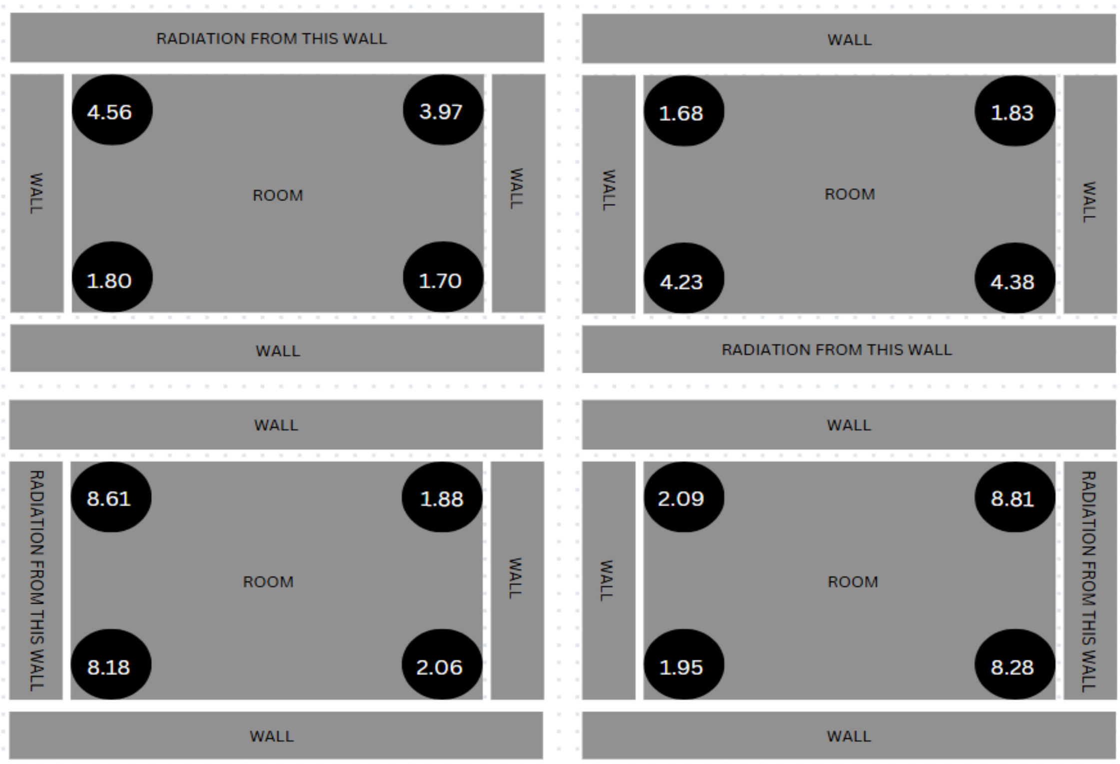

Figure 5.5. A schematic view of simulations done near the four corners of the room and the dose received from the 1.46 MeV gamma energy of ^{40}K in the different walls in units of 10^{-13} Gy.

This simulation was done to first test whether walls of similar dimensions provided similar doses due to the presence of similar compositions in them. It also investigated the wall and corners of the room which provided the most dose to the person in the room, relative to the radiation dose received from that wall. The simulations capitalised on the one geometry beam definition capability of TOPAS, allowing the radiation to be defined separately in all four walls, ceiling and floor, and enabling the visualization of the dose contributed from each of them at different points of the room.

When the gamma energies from the two big walls were simulated, dose received by the individual for being closer to them were recorded averagely as 4.26 x 10^{-13} and 4.31 x 10^{-13} Gy respectively. When the individual was positioned near the opposite big wall, while the same simulation is undertaken, the average dose received were 1.75 x 10^{-13} and 1.76 x 10^{-13} Gy respectively. The dose received was high when the individual was closer to the gamma emitting walls than when far away. Due to the similarity in doses received at positions near and far away from the gamma emitting wall, it was proven that walls of similar dimensions contributed similar doses to the individual due to the presence of similar composition and distribution of the gamma energies in the wall. The average dose received near the two smaller gamma emitting walls were 8.40 x 10^{-13} and 8.55 x 10^{-13} Gy respectively and the dose received from the position opposite them were 1.97 x 10^{-13} and 2.02 x 10^{-13} Gy respectively. Comparison of the small wall simulations to the simulations made from the big walls showed twice as high doses from the small walls when the individual was closer to them than the big walls. This can be attributed to the high energy distribution and self-absorption in the bigger walls than the smaller walls. Dose measured from the floor and ceiling for all positions in the room were constant with an average of 6.08 x 10^{-13} and 3.14 x 10^{-13} Gy respectively. The dose from the floor was higher because the position of the individual was defined closer to it to realistically depict a person standing in the room. This simulation further confirms that most doses at a particular point in a room is mainly from the wall the individual is close to.

5.3. Simulations at the centre of the room and different room sizes

5.3.1. Centre of the room

Simulations were performed with the individual positioned in the centre of the room. This was chosen because it meant equal amount of energy were detected from walls of similar dimensions, as demonstrated in section 5.2.2. When compared to the simulations near the walls, the dose received at the centre was less than the dose received near the walls. For instance, the dose for one of the big walls was 4.56 x 10^{-13} Gy near the wall and 3.75 x 10^{-13} Gy at the centre of the room for 1.46 MeV. Figure 5.6 shows the dose obtained from the gamma energies of the radionuclides from the walls at the centre of the room. The total dose received by the individual at the centre of the room is 2.39 x 10^{-11} Gy (Simulated at 100,000 events per geometry for 6 geometries). The small walls contributed the most dose to the individual than the big wall, just like the near wall simulations, and it was consistent for all the energies, with ^{40}K contributing more dose than the other radionuclides. Dose from the floor also increased as the gamma energy increased.

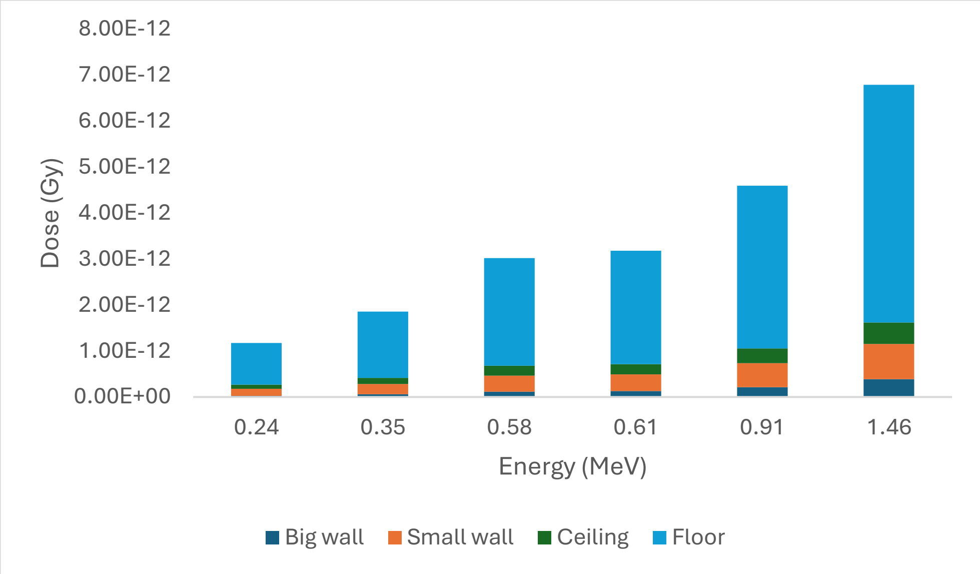

Figure 5.6. Dose contribution of walls, ceiling and floor to the individual at the centre of the room from different energies for 600000 events (100000 per geometry component) (0.24 for ^{212}Pb, 0.35 for ^{214}Pb, 0.58 for ^{208}Tl, 0.61 for ^{214}Bi, 0.91 for ^{228}Ac and 1.46 for ^{40}K (All energies in MeV)).

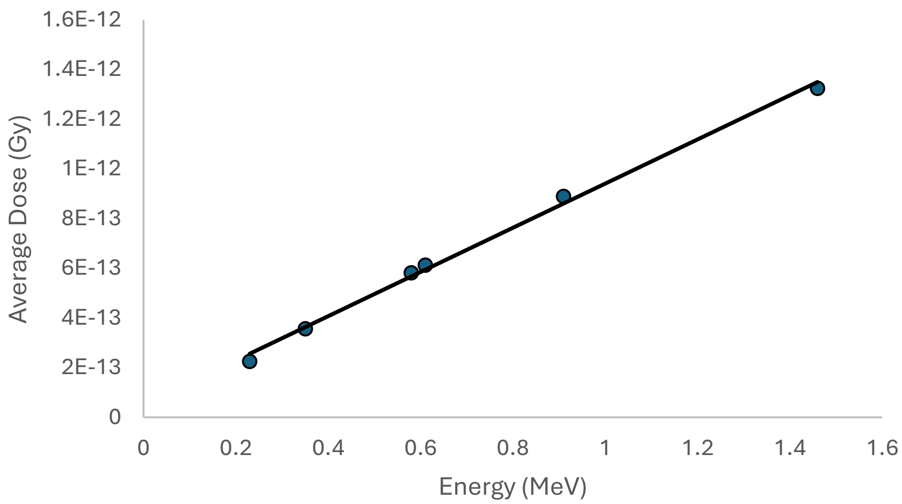

Figure 5.7. Dose-energy graph that shows the average dose contributed by the gamma energies of the radionuclide to the individual at the centre of the room.

The average dose received from the entirety of the walls, floors and ceilings from the radionuclides is presented in figure 5.7. Lower energies like 0.24 MeV for ^{212}Pb and 0.35 MeV for ^{214}Pb contributed low doses of 2.25 x 10^{-13} and 3.56 x 10^{-13} Gy respectively. Similar gamma energies like 0.58 MeV for ^{208}Tl and 0.61 MeV for ^{214}Bi contributed 5.82 x 10^{-13} and 6.12 x 10^{-13} Gy respectively and high energies like 0.91 MeV for ^{228}Ac and 1.46 MeV for ^{40}K produced higher doses of 8.89 x 10^{-13} and 1.32 x 10^{-12} Gy respectively. The relationship between them was linear with higher energies contributing high doses and vice versa.

5.3.2. Different room sizes and contribution from the terrestrial natural radiation

Different room sizes were defined in the code to investigate their effect on the dose received by the individual. The room sizes investigated were 4 x 2.5 x 3 m^3 for a small house, the model house (5 X 2.8 X 4 m^3) and 8 x 2.8 x 4 m^3 for a large house (Orabi, 2016). The thickness for the walls, ceiling and floor were all 20 cm for all the houses.

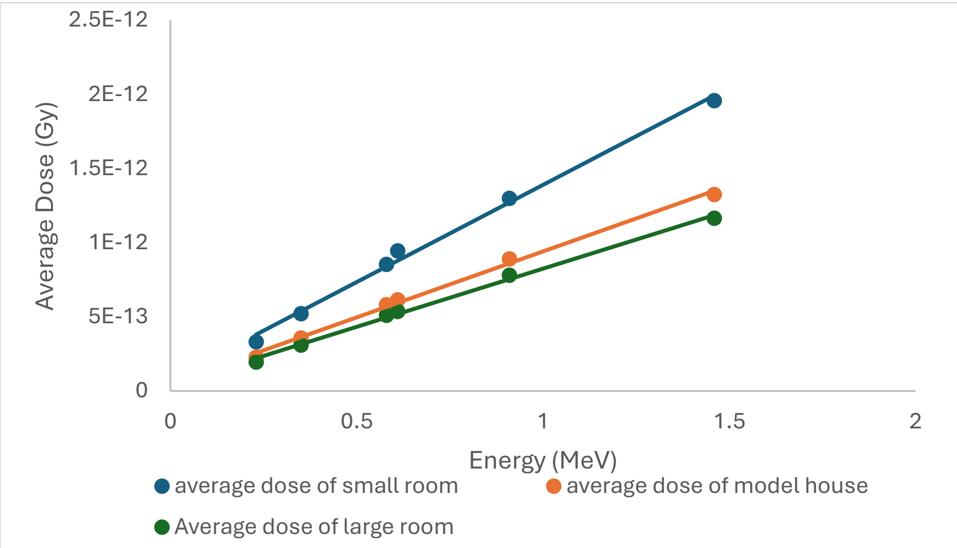

Figure 5.8. Dose-energy graph that shows the average dose contributed by the energies of the radionuclide to the individual at the centre of the room for three different room sizes.

The individual was positioned at the centre of the room for all the simulations. The simulations entailed the energies listed in Table 4.1. The results of the simulations are shown in figure 5.8. Total dose from the three houses were 3.54 (small), 2.39 (model) and 2.09 (large) (all in units of 10^{-11} Gy). Dose from the energies in the smaller house were higher than the other houses. This is possible because the distance between the individual and the walls in the smaller house were reduced, hence, the gamma energies did not have to travel far to reach the individual and more were able to interact with the individual. Although the model house showed high dose contribution to the individual than the large house, the difference between them was not as significant as the dose contribution from the small house. This can be attributed to the energies having to travel longer distances to reach the individual, hence more lose their energy. An interesting example is noted in the dose contribution of the high energy 1.46 MeV of ^{40}K in the model house and the large house. As discussed earlier in figure 5.6, the small walls contributed more dose than the big ones due to high distribution and self-absorption in the big walls. This was also evident for the small house but for the large house, the dose contributed for the small and big walls were similar, with an average of 3.68 x 10^{-13} Gy for the big wall and 3.08 x 10^{-13} Gy for the small walls. This shows that distance plays a vital role in dose contribution to an individual.

Finally, the average dose contribution from the three terrestrial radionuclides (^{232}Th, ^{238}U and K) were computed from finding the average dose of decay products that contributes to that radionuclide. This was done for all three houses and is presented in table 5.2.

		Average Dose (Gy)		
Nuclide	**Energy**	**Small House**	**Model House**	**Large House**
^{40}K	1.46	1.96E-12	1.32E-12	1.16E-12
^{238}U-Series				
^{214}Pb	0.35	7.31E-13	4.84E-13	4.21E-13
^{214}Bi	0.61			
^{232}Th-series				
^{212}Pb	0.24	8.26E-13	5.65E-13	4.94E-13
^{208}Tl	0.58			
^{228}Ac	0.91			

Table 5.2. Average dose contribution from the terrestrial radionuclides to the individual in all three house sizes.

The most dose was contributed by ^{40}K, followed by the decay series of ^{232}Th and finally, the decay series of ^{238}U. This result is consistent with theory that, ^{40}K has the highest composition in walls and hence contributes the most dose in that case, however, they do not necessarily cause complications to the individual like the decay products of ^{238}U and ^{232}Th (Cinelli et al., 2019).

6. CONCLUSION

The dose received by an individual in a room from terrestrial natural background radiation (^{238}U, ^{232}Th and ^{40}K) was investigated using simulations developed from a code created using the GEANT4 Monte-Carlo toolkit, Tool for Particle Simulation (TOPAS). The simulations were done for a model house with room dimensions of 5 X 2.8 X 4 m^3 and walls, ceiling and floor made of concrete with thickness 20 cm. The gamma energy of the natural background radiation was emitted from the concrete defined walls, ceiling and floor into the room.

To validate the defined concrete material for the simulation, a simple set-up of a concrete wall placed between a germanium detector and a point source was defined in a separate TOPAS code. This was used to investigate the attenuation coefficient of the concrete material for different gamma energies. The concrete demonstrated similar values closer to the attenuation coefficient of the gamma energies obtained from a national standard database.

In the indoor simulations, near wall simulations proved that being close to a radiation emitting wall will expose you to higher doses than being far away. the thickness of one of the big walls was increased while the individual was at the centre of the room. This was done to investigate increasing thickness on the dose received by the individual (practically increasing radionuclide composition, more gamma energies). Two energies were investigated in this simulation, the high energy (1.46 MeV) of ^{40}K and the low energy (0.24) of ^{212}Pb. Low energies contributed low doses which increased and rapidly declined with increasing thickness in comparison to high energies. This happened due to the walls acting as source and shield to the radiation being emitted from it.

Finally, simulations at the centre of the room proved that small walls contributed high doses than big walls due to less distribution and self-absorption. Another simulation comparing three room dimensions proved that dose received in smaller houses are higher than dose received in big houses. The less dose in big houses resulted from the gamma energy fall off as they travel longer distances to reach the individual in the centre of the room and more self-absorption in the walls. The dose contribution of the three terrestrial radionuclides were investigated in the small, model and big houses. ^{40}K was noted to contribute the most dose among the radionuclides, ^{232}Th was the second highest and ^{238}U contributed the least.

7. FUTURE WORKS

The study was limited by time. However, an extensive reach was achieved on the factors affecting indoor natural background radiation. It is recommended that future works focus on increasing the number of events simulated through the walls to millions to investigate how that affects the simulations. Windows and doors could also be added to the house to see if they affect the dose received. Inner wall partitions are also recommended in further studies to further analyse if they reduce or increase the dose. Absorbers could also be added to the wall to investigate how much they reduce the dose. Furthermore, two or more houses could be investigated to see how adjacent houses affect the dose received by the individual. Finally, experimental designs for investigating the terrestrial radionuclides in buildings could factor simulations from this code as a comparison metric.

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10. APPENDIX

A.1. TOPAS CODE FOR SIMULATION

Below is the TOPAS code used for the simulations

Note that '#' is the comment syntax for TOPAS, everywhere it is seen is a comment guide

'#' can also be used to suspend the functionality of a code

DEFINE WORLD

#This section of the code specifies the world of the simulation.

#In TOPAS, a world definition is required to create a space in which the simulations will take place.

#The geometry of the world serves as a parent for the most significant geometries defined in the code. The world is defined below

s:Ge/World/Type = "TsBox" #This defines the type of geometry of the world. In this code, it was specified as a box

s:Ge/World/Material = "G4_AIR" #The World was made of air to simulate a realistic environment for the code

d:Ge/World/HLX = 5 m #The next three lines are the dimensions of the world, it is reported in half lengths, actual value is 10 m

d:Ge/World/HLY = 5 m

d:Ge/World/HLZ = 5 m

Beam source profile

#The gamma energies of the terrestrial radionuclides were defined here

s:So/MyBeam/Type = "Beam" #Defines a beam of primary particles s:So/MyBeam/Component = "FLOOR" #Location of the beam, other components used were the concrete walls, defined as "WALL1", "WALL2", "WALL3", WALL4" and the "CEILING"

s:So/MyBeam/BeamParticle = "gamma" #Simulations were for photons so gamma beam particle was defined.

d:So/MyBeam/BeamEnergy = 1.46 MeV #Energy of the terrestrial radionuclides (Other energies are presented in table 4.1)

u:So/MyBeam/BeamEnergySpread = 0.01 # Beam standard deviation in %

s:So/MyBeam/BeamPositionCutoffShape = "Rectangle" #Automatically, a shape for beam cut off is supposed to be defined. In this case, it was rectangular because it was from the walls

d:So/MyBeam/BeamPositionCutoffX = 2.0 m #This ensured that the beam was emitted from everywhere in the wall in the x direction. A small value will make the beam a point source

d:So/MyBeam/BeamPositionCutoffY = 2.0 m #Same as the CutOffX but in the y direction

s:So/MyBeam/BeamPositionDistribution = "Flat" #The distribution was uniform

d:So/MyBeam/BeamAngularCutoffX = 180. deg #This ensured a maximum beam spread in the x direction

d:So/MyBeam/BeamAngularCutoffY = 180. deg #maximum beam spread in the y direction

s:So/MyBeam/BeamAngularDistribution = "Flat" #The angular distribution is also uniform

BUILDING

#This part of the code developed the building

#HLX, HLY, HLZ represent half lenth of the dimensions of the components in x, y and z respectively

#Current parameters in the code are for the big house, the other two house's dimensions have been commented at their respective Half lengths

INNER GEOMETRY FILLED WITH AIR

s:Ge/ROOM/Type = "TsBox" #Geometry of room

```
s:Ge/ROOM/Parent = "World" #Room is located in the World
s:Ge/ROOM/Material = "G4_AIR" #The inside of the room was also filled with air
d:Ge/ROOM/TransZ = 0. m #Located in the middle of the world
d:Ge/ROOM/HLX = 4 m # 2.5 m for model house, 2 m for small house (Length)
d:Ge/ROOM/HLY = 2 m # 2. m for model house, 1.5 m for small house (Height)
d:Ge/ROOM/HLZ = 1.4 m #1.4 m for model house, 1.25 m for small house (Width)
s:Ge/ROOM/Color = "white" #color of the room


#### CONCRETE WALLS ####
######## BIG WALL1 #######
s:Ge/WALL1/Type = "TsBox" #First big wall on the left
s:Ge/WALL1/Material = "G4_CONCRETE" #The wall was made of concrete
s:Ge/WALL1/Parent = "World"
dc:Ge/WALL1/HLX = 4.2 m #2.7 m for model house, 2.2 m for small house
dc:Ge/WALL1/HLY = 2. m # 2 m for model house, 1.5 m for small house
dc:Ge/WALL1/HLZ = 10 cm # Thickness of wall, it's a half length for 20cm and was maintained throughout the simulation
dc:Ge/WALL1/TransX = 0. m
dc:Ge/WALL1/TransY = 0. m
dc:Ge/WALL1/TransZ = 1.6 m # 1.5 m for model house, 1.35 m for small house (Position in the world)
d:Ge/WALL1/RotX = 0. deg
d:Ge/WALL1/RotY = 0. deg
d:Ge/WALL1/RotZ = 0. deg
```

i:Ge/WALL1/XBins = 1

i:Ge/WALL1/YBins = 1

i:Ge/WALL1/ZBins = 1

s:Ge/WALL1/Color = "blue"

s:Ge/WALL1/DrawingStyle = "WireFrame" #Drawing style was wireframe to make it transparent. for opaque, use "solid"

#Big wall2 is a reflection of big wall1 in the opposite direction, TransZ makes that possible

BIG WALL2

s:Ge/WALL2/Type = "TsBox"

s:Ge/WALL2/Material = "G4_CONCRETE"

s:Ge/WALL2/Parent = "World"

d:Ge/WALL2/HLX = 4.2 m

d:Ge/WALL2/HLY = 2. m

dc:Ge/WALL2/HLZ = 10 cm # This was maintained throughout the simulation

d:Ge/WALL2/TransX = 0. m

d:Ge/WALL2/TransY = 0. m

d:Ge/WALL2/TransZ = -1.6 m #Positions the wall at an exact opposite point as BIG WALL1 in the World

d:Ge/WALL2/RotX = 0. deg #The wall was not rotated

d:Ge/WALL2/RotY = 0. deg

d:Ge/WALL2/RotZ = 0. deg

i:Ge/WALL2/XBins = 1

i:Ge/WALL2/YBins = 1

i:Ge/WALL2/ZBins = 1

s:Ge/WALL2/Color = "blue"

s:Ge/WALL2/DrawingStyle = "WireFrame"

#Just as TransZ manipulates the position for the big walls, TransX does the same for the small walls

######## SMALL WALL 3 #######

s:Ge/WALL3/Type = "TsBox" #Left part of wall

s:Ge/WALL3/Material = "G4_CONCRETE" #"G4_WATER" ## Options: "Lead" "G4_W"

s:Ge/WALL3/Parent = "World"

d:Ge/WALL3/HLX = 10 cm # Was unchanged the entire simulation

d:Ge/WALL3/HLY = 2 m #2 m for model house, 1.5 m for small house

d:Ge/WALL3/HLZ = 1.4 m #1.4 m for model house, 1.25 for small house

d:Ge/WALL3/TransX = 4.1 m #2.6 m for model house, 2.15 m for small house

d:Ge/WALL3/TransY = 0. m

d:Ge/WALL3/TransZ = 0. m

d:Ge/WALL3/RotX = 0. deg

d:Ge/WALL3/RotY = 0. deg

d:Ge/WALL3/RotZ = 0. deg

i:Ge/WALL3/XBins = 1

i:Ge/WALL3/YBins = 1

i:Ge/WALL3/ZBins = 1

s:Ge/WALL3/Color = "blue"

s:Ge/WALL3/DrawingStyle = "WireFrame"

SMALL WALL 4

s:Ge/WALL4/Type = "TsBox" #Left part of wall

s:Ge/WALL4/Material = "G4_CONCRETE" #"G4_WATER" ## Options: "Lead" "G4_W"

s:Ge/WALL4/Parent = "World"

d:Ge/WALL4/HLX = 10 cm

d:Ge/WALL4/HLY = 2 m

d:Ge/WALL4/HLZ = 1.4 m

d:Ge/WALL4/TransX = -4.1 m

d:Ge/WALL4/TransY = 0. m

d:Ge/WALL4/TransZ = 0. m

d:Ge/WALL4/RotX = 0. deg

d:Ge/WALL4/RotY = 0. deg

d:Ge/WALL4/RotZ = 0. deg

i:Ge/WALL4/XBins = 1

i:Ge/WALL4/YBins = 1

i:Ge/WALL4/ZBins = 1

s:Ge/WALL4/Color = "blue"

s:Ge/WALL4/DrawingStyle = "WireFrame"

#TransY changes the position for the floor and ceiling

CEILING

s:Ge/CEILING/Type = "TsBox" #Left part of wall

s:Ge/CEILING/Material = "G4_CONCRETE" #"G4_WATER" ## Options: "Lead" "G4_W"

```
s:Ge/CEILING/Parent               = "World"
d:Ge/CEILING/HLX                  = 4.2 m #2.7 m for model house, 2.2 m for small house
d:Ge/CEILING/HLY                  = 10. cm # remained unchanged
d:Ge/CEILING/HLZ                  = 1.6 m #1.6 m for model house, 1.45 m for small
house
d:Ge/CEILING/TransX               = 0 m
d:Ge/CEILING/TransY               = 2.1 m #2.1 m for model house, 1.6 m for small house
d:Ge/CEILING/TransZ               = 0. m
d:Ge/CEILING/RotX                 = 0. deg
d:Ge/CEILING/RotY                 = 0. deg
d:Ge/CEILING/RotZ                 = 0. deg
i:Ge/CEILING/XBins                = 1
i:Ge/CEILING/YBins                = 1
i:Ge/CEILING/ZBins                = 1
s:Ge/CEILING/Color                = "green"
s:Ge/CEILING/DrawingStyle      = "wireframe"


#### FLOOR #####
s:Ge/FLOOR/Type                = "TsBox" #Left part of wall
s:Ge/FLOOR/Material               = "G4_CONCRETE" #"G4_WATER" ## Options:
"Lead" "G4_W"
s:Ge/FLOOR/Parent                 = "World"
d:Ge/FLOOR/HLX                 = 4.2 m
d:Ge/FLOOR/HLY                 = 10. cm
d:Ge/FLOOR/HLZ                 = 1.6 m
d:Ge/FLOOR/TransX                 = 0 m
```

d:Ge/FLOOR/TransY = -2.1 m

d:Ge/FLOOR/TransZ = 0. m

d:Ge/FLOOR/RotX = 0. deg

d:Ge/FLOOR/RotY = 0. deg

d:Ge/FLOOR/RotZ = 0. deg

i:Ge/FLOOR/XBins = 1

i:Ge/FLOOR/YBins = 1

i:Ge/FLOOR/ZBins = 1

s:Ge/FLOOR/Color = "yellow"

s:Ge/FLOOR/DrawingStyle = "wireframe"

DETECTOR (INDIVIDUAL)

s:Ge/MyDetector/Type = "TsCylinder" #The detector was a cylinder

s:Ge/MyDetector/Material = "G4_WATER" #The cylinder was filled with water

s:Ge/MyDetector/Parent = "ROOM" #It was located in the room

d:Ge/MyDetector/RMin = 10 cm #The minimum radius

d:Ge/MyDetector/RMax = 15 cm #The maximum radius

d:Ge/MyDetector/HL = 0.9 m #Height of the detector

d:Ge/MyDetector/SPhi = 0 deg #Sphi rotation

d:Ge/MyDetector/DPhi = 360 deg #DPhi rotation

dc:Ge/MyDetector/TransX = 0 m

dc:Ge/MyDetector/TransY = -1 m # -1 m for model house, -0.6 m for small house (position in the room)

dc:Ge/MyDetector/TransZ = 0 m

d:Ge/MyDetector/RotX = 90. deg #Rotation was done to make the detector stand upright

d:Ge/MyDetector/RotY = 0. deg

d:Ge/MyDetector/RotZ = 0. deg

i:Ge/MyDetector/XBins = 1

i:Ge/MyDetector/YBins = 1

i:Ge/MyDetector/ZBins = 1

s:Ge/MyDetector/Color = "red"

s:Ge/MyDetector/DrawingStyle = "Solid"

sv:Ph/Default/Modules = 1 "g4em-standard_opt0"

SCORERS

s:Sc/MyDetectorEnergy/Quantity = “DoseToMedium” # type of data recorded, energy deposited per unit mass of medium, Option: EnergyDeposit

s:Sc/MyDetectorEnergy/Component = “MyDetector” # the subject of the score

b:Sc/MyDetectorEnergy/OutputToConsole = “TRUE”

b:Sc/MyDetectorEnergy/PropagateToChildren = “True”

b:Sc/MyDetectorEnergy/OutputAfterRun = “True”

s:Sc/MyDetectorEnergy/OutputFile = “Energy”

s:Sc/MyDetectorEnergy/OutputType = “csv” # Options: “csv” “Root” “Xml”

s:Sc/MyDetectorEnergy/IfOutputFileAlreadyExists = “Overwrite”

sv:Sc/MyDetectorEnergy/Report = 2 “Sum” “Standard_Deviation” ## Dose calculation that was scored

GRAPHIC WINDOW AND Qt

#TOPAS allows visualization of simulations but for accurate results at higher number of histories, the visualizations are to be commented out

#Comment out the two lines that have been referred to below to turn off the visualization

```
s:Gr/ViewA/Type                    = "OpenGL" # comment out this line...
i:Gr/ViewA/WindowSizeX             = 900
i:Gr/ViewA/WindowSizeY             = 900
d:Gr/ViewA/Theta                   = 45 deg
d:Gr/ViewA/Phi                     = 45 deg
d:Gr/ViewA/PerspectiveAngle        = 30 deg
u:Gr/ViewA/Zoom                    = 3
Ts/UseQt = "True"      # ...and this line to turn off vis for quicker runtime


i:So/MyBeam/NumberOfHistoriesInRun   =   100000          # how many particles/ events


Ge/QuitIfOverlapDetected   = "True"
```

ATTENUATION TOPAS CODE

```
##### DEFINE WORLD ############


s:Ge/World/Type = "TsBox"

s:Ge/World/Material = "Air" # "Vacuum"

d:Ge/World/HLX = 5 m # Half lengths

d:Ge/World/HLY = 5 m

d:Ge/World/HLZ = 5 m




##### BEAM ########################

#### Physical object ####

s:Ge/MyBeamPosition/Type = "TsSphere" #"Group"

s:Ge/MyBeamPosition/Parent = "World"

s:Ge/MyBeamPosition/Material     = "Air" #"G4_WATER"

d:Ge/MyBeamPosition/RMin        = 0 m       # Radius

d:Ge/MyBeamPosition/RMax        = 0.05 m

d:Ge/MyBeamPosition/SPhi        = 0. deg    # Start phi

d:Ge/MyBeamPosition/DPhi        = 360. deg          # Phi range covered

d:Ge/MyBeamPosition/STheta      = 0. deg            # Start theta

d:Ge/MyBeamPosition/DTheta      = 180. deg          # Theta range covered

d:Ge/MyBeamPosition/TransX      = 0. m     # Translations

d:Ge/MyBeamPosition/TransY      = 0. m

d:Ge/MyBeamPosition/TransZ      = -1 m

d:Ge/MyBeamPosition/RotX        = 0. deg # Rotations done from PoV along axis, clockwise

d:Ge/MyBeamPosition/RotY        = 0. deg
```

```
d:Ge/MyBeamPosition/RotZ         = 0. deg

s:Ge/MyBeamPosition/DrawingStyle = "WireFrame"   # Options: "WireFrame" "Solid"

s:Ge/MyBeamPosition/Color            = "white"        # Options: most colours but use lower-
case



#### Particle source parameters ####

s:So/MyBeam/Type                    = "Beam"

s:So/MyBeam/Component              = "MyBeamPosition"

s:So/MyBeam/BeamParticle           = "gamma"



d:So/MyBeam/BeamEnergy             = 1.46 MeV     # will take MeV, GeV as units

u:So/MyBeam/BeamEnergySpread        = 0.01        # in %

s:So/MyBeam/BeamPositionCutoffShape  = "Rectangle"

d:So/MyBeam/BeamPositionCutoffX      = 0.05 m

d:So/MyBeam/BeamPositionCutoffY      = 0.05 m

s:So/MyBeam/BeamPositionDistribution = "Flat"

d:So/MyBeam/BeamAngularCutoffX       = 1. deg

d:So/MyBeam/BeamAngularCutoffY       = 1. deg

s:So/MyBeam/BeamAngularDistribution  = "Flat"
```

```
#### PHYSICS OF SIMULATION ####

sv:Ph/Default/Modules = 1 "g4em-standard_opt0" # options: "g4em-standard_optX" where X=0,1,2,3,4 "g4radioactivedecay"

#### GEOMETRY ####
#### SLAB OF MATERIAL #######

s:Ge/SLAB/Type     = "TsBox"
s:Ge/SLAB/Material = "G4_CONCRETE"
s:Ge/SLAB/Parent   = "World"
d:Ge/SLAB/HLX      = 0.25 m
d:Ge/SLAB/HLY      = 0.25 m
dc:Ge/SLAB/HLZ      = 10 cm
d:Ge/SLAB/TransX   = 0. m
d:Ge/SLAB/TransY   = 0. m
d:Ge/SLAB/TransZ   = -0.3 m
d:Ge/SLAB/RotX     = 0. deg
d:Ge/SLAB/RotY     = 0. deg
d:Ge/SLAB/RotZ     = 0. deg
i:Ge/SLAB/XBins    = 1
i:Ge/SLAB/YBins    = 1
i:Ge/SLAB/ZBins    = 1
s:Ge/SLAB/Color    = "blue"
s:Ge/SLAB/DrawingStyle = "WireFrame" # options: WireFrame, Solid
```

```
#### DETECTOR #######

s:Ge/MyDetector/Type     = "TsBox"
s:Ge/MyDetector/Material = "G4_Ge" #"G4_WATER" ## Options: "Lead" "G4_W"
s:Ge/MyDetector/Parent   = "World"
d:Ge/MyDetector/HLX      = 0.25 m
d:Ge/MyDetector/HLY      = 0.25 m
d:Ge/MyDetector/HLZ      = 10 cm
d:Ge/MyDetector/TransX   = 0. m
d:Ge/MyDetector/TransY   = 0. m
d:Ge/MyDetector/TransZ   = 0.3 m
d:Ge/MyDetector/RotX     = 0. deg
d:Ge/MyDetector/RotY     = 0. deg
d:Ge/MyDetector/RotZ     = 0. deg
i:Ge/MyDetector/XBins    = 1
i:Ge/MyDetector/YBins    = 1
i:Ge/MyDetector/ZBins    = 1
s:Ge/MyDetector/Color    = "red"
s:Ge/MyDetector/DrawingStyle = "WireFrame"
```

SCORERS

#In this code, two scorers were defined, the first was to score the energy after interaction (EnergyDeposit)

#A useful scorer for calculating attenuation

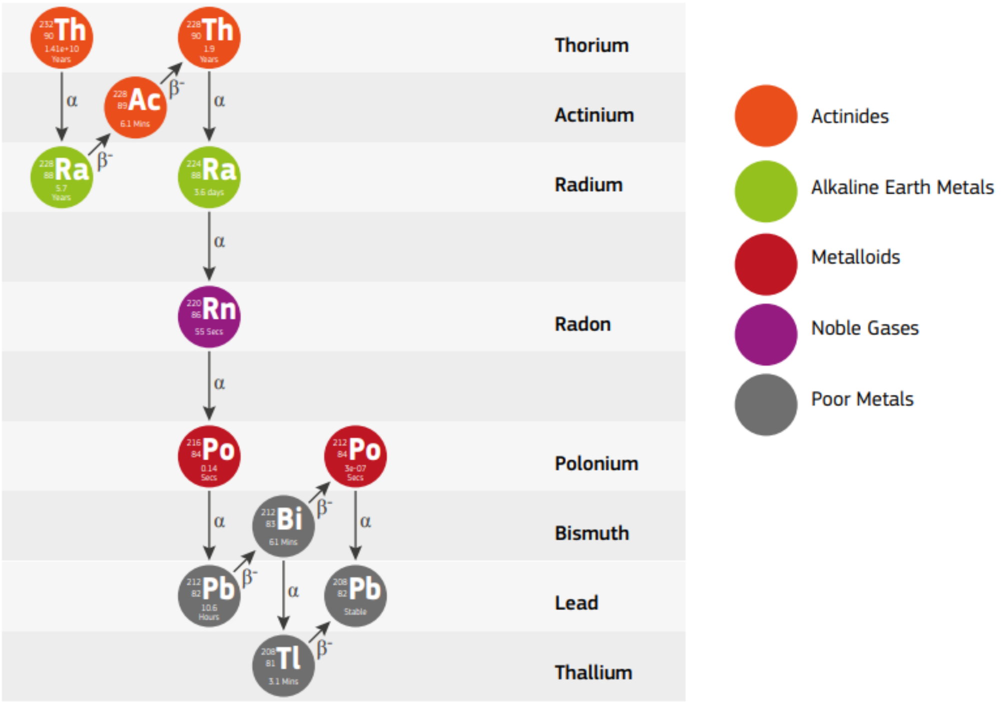
#The second scorer was IncidentTrack and it recorded the counts of the energy in a detector for the gamma spectrum of ^{40}K

#Note: To use either scorer, the other must be commented out

```
s:Sc/MyEnergy/Quantity                    = "EnergyDeposit" # type of data recorded



s:Sc/MyEnergy/Component                   = "MyDetector"    # the subject of the score
b:Sc/MyEnergy/OutputToConsole             = "TRUE"
s:Sc/MyEnergy/OutputFile                  = "Energy"
s:Sc/MyEnergy/OutputType                  = "csv"    # Options: "csv" "Root" "Xml"
s:Sc/MyEnergy/IfOutputFileAlreadyExists   = "Overwrite"
sv:Sc/MyEnergy/Report = 2 "Sum" "Standard_Deviation" ## Options: "Mean" "Variance"

#s:Sc/MyEnergy/Quantity                = "EnergyDeposit"
#s:Sc/MyEnergy/Component               = "MyDetector"
#b:Sc/MyEnergy/OutputToConsole         = "TRUE"
#i:Sc/MyEnergy/EBins                   = 200
#d:Sc/MyEnergy/EBinMin                 = 0. MeV
#d:Sc/MyEnergy/EBinMax                 = 2. MeV
#sc:Sc/MyEnergy/EBinEnergy             = "IncidentTrack" #
```

```
#sv:Sc/MyEnergy/Report = 1 "Count_In_Bin"



##### GRAPHIC WINDOW AND Qt ################


#s:Gr/ViewA/Type                    = "OpenGL" # comment out this line...

i:Gr/ViewA/WindowSizeX              = 900

i:Gr/ViewA/WindowSizeY              = 900

d:Gr/ViewA/Theta                    = 45 deg

d:Gr/ViewA/Phi                      = 45 deg

d:Gr/ViewA/PerspectiveAngle         = 30 deg

u:Gr/ViewA/Zoom                     = 3



#Ts/UseQt = "True"     # ...and this line to turn off vis for quicker runtime

#b:Gr/ViewA/Active = "False"


i:So/MyBeam/NumberOfHistoriesInRun   = 100000    # how many particles
```

A.2. URANIUM AND THORIUM DECAY CHAINS



Radioactive decay chain of ^{232}Th (Cinelli et al., 2019).

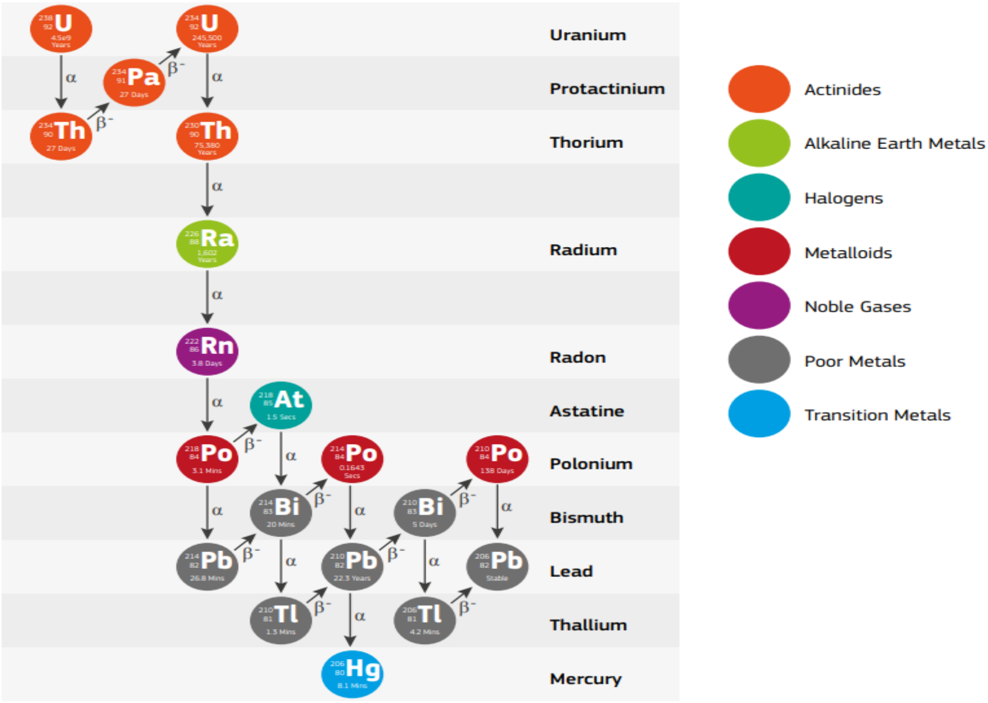


Figure 2.7. Radioactive decay chain of ^{238}U (Cinelli et al., 2019).

A.3. NEAR WALL SIMULATIONS FOR OTHER RADIONUCLIDES

^{214}Pb	**Big Wall 1**	**Big Wall 2**	**Small Wall 1**	**Small Wall 2**	**Ceiling**	**Floor**
CORNER 1 (Big Wall 2, Small Wall 2)	3.07E-14	5.77E-14	6.35E-14	2.37E-13	9.28E-14	1.82E-13
CORNER 2 (Big Wall 2, Small Wall 1)	2.80E-14	4.93E-14	2.53E-13	7.00E-14	9.56E-14	2.07E-13
CORNER 3 (Big Wall 1, Small Wall 1)	6.06E-14	2.92E-14	2.39E-13	5.92E-14	8.85E-14	1.80E-13
CORNER 4 (Big Wall 1, Small Wall 2)	5.81E-14	3.01E-14	6.71E-14	2.46E-13	9.07E-14	1.96E-13

Dose recorded for the gamma energy (0.35 MeV) of ^{214}Pb near the walls in Gy for 100000 events in each wall/ ceiling/ floor (The Corners represent where the cylinder is and the Walls, ceiling and floor represent where the radiation source is emitted from).

^{208}Tl	Big Wall 1	Big Wall 2	Small Wall 1	Small Wall 2	Ceiling	Floor
CORNER 1 (Big Wall 2, Small Wall 2)	5.44E-14	1.20E-13	9.44E-14	3.77E-13	1.39E-13	2.82E-13
CORNER 2 (Big Wall 2, Small Wall 1)	5.76E-14	1.14E-13	3.99E-13	1.03E-13	1.44E-13	3.06E-13
CORNER 3 (Big Wall 1, Small Wall 1)	1.28E-13	5.93E-14	3.79E-13	8.24E-14	1.35E-13	2.96E-13
CORNER 4 (Big Wall 1, Small Wall 2)	1.28E-13	6.25E-14	9.82E-14	3.86E-13	1.46E-13	3.03E-13

Table 5.4. Dose recorded for the gamma energy (0.58) of ^{208}Tl near the walls in Gy (The Corners represent where the cylinder is and the Walls, ceiling and floor represent where the radiation source is emitted from).